

Nevada Historical and Environmental Geospatial Repository

Technical Architecture and Sustainability Roadmap

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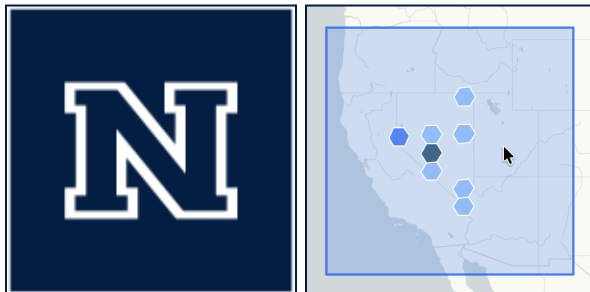
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Building on a Legacy of Connecting People to Place: A Strategic Plan for Nevada's Geospatial Repository

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Executive Summary

Section Summary

UNR can launch a sustainable statewide geoportal without purchasing or maintaining a large proprietary platform. This report recommends a structured pilot using the community-sponsored OpenGeoMetadata API, a lightweight customizable frontend, standardized Aardvark metadata, and cloud-optimized assets in low-cost object storage. This open-source approach is achievable, cost-effective, and portable—but its long-term success still requires named service ownership, metadata stewardship, and dependable staff support.

Recommendation

UNR should develop its statewide geoportal as an open-source, standards-based service in partnership with the OpenGeoMetadata community. The recommended architecture combines OpenGeoMetadata Aardvark records, cloud-optimized assets, the community-sponsored OpenGeoMetadata API, and a customizable React frontend. Original files should remain under established preservation controls, while Cloud Optimized GeoTIFFs, PMTiles, IIIF images, and related derivatives support public previews and downloads.

UNR should begin with the OpenGeoMetadata API and custom frontend demonstrated by the existing prototype. GeoBlacklight should remain the established alternative if the newer application stack does not meet UNR's requirements for reliability, accessibility, maintainability, governance, or support.

Reasoning

This approach offers the strongest balance of affordability, sustainability, interoperability, and institutional control. It avoids recurring platform-license costs and minimizes the local infrastructure required to operate the service. Aardvark metadata can be maintained through GitHub and OpenGeoMetadata Studio; the public frontend can be hosted through static web hosting services (like GitHub Pages); and cloud-optimized assets can reside in comparatively inexpensive S3-compatible object storage. The principal long-term costs are therefore staff stewardship, asset storage, quality assurance, and service support—not software licensing or a large server environment.

The recommended architecture also protects UNR's investment. Standards-based metadata and cloud-optimized files remain portable if the public interface changes, allowing the same resources to support GeoBlacklight, GIS applications, digital exhibits, teaching, and future

research tools. UNR can pursue this model with a working five-year total-cost-of-ownership target of approximately **\$300,000 including contingency**, subject to validation during implementation planning.

Next Steps

UNR should undertake a defined pilot phase using representative collections and the existing prototype. The pilot should:

- Validate the complete workflow from metadata creation and asset processing through indexing, discovery, previews, and downloads.
- Test accessibility, security, monitoring, backups, analytics, correction procedures, and realistic operating costs.
- Establish service ownership, staffing responsibilities, governance, collection-selection criteria, and annual operating commitments.
- Engage campus IT, preservation staff, collection owners, statewide partners, Tribal Nations, and other represented communities as appropriate.
- Evaluate the OpenGeoMetadata API against agreed production-readiness benchmarks while preserving GeoBlacklight as a viable fallback.

A production launch should proceed once UNR has confirmed that the service can be operated reliably, accessibly, and sustainably. The technical foundation is already within reach; the remaining leadership task is to establish the ownership, governance, and continuing support needed to sustain it.

Introduction

Section Summary

This report translates UNR's grant objectives and statement of work into a practical plan for launching a statewide historical and environmental geoportal. It evaluates metadata, digital assets, platform options, costs, web practices, and a working prototype, ultimately recommending an open-source architecture built around OpenGeoMetadata standards and cloud-optimized delivery. Eric Larson was engaged for his experience with academic geospatial repositories, GeoBlacklight, OpenGeoMetadata, metadata lifecycle systems, and the Big Ten Academic Alliance's emerging API and frontend architecture.

Nevada's historical maps, aerial imagery, land records, environmental datasets, and related archival materials document the relationship between people and place across a state defined by distance, environmental change, and uneven access to cultural infrastructure. These resources support scholarship, teaching, public history, land management, journalism, genealogy, Indigenous history, and civic decision-making. Yet many remain distributed across institutional collections, agency systems, archival storage, and local files. Some are technically preserved but difficult for the public to discover, interpret, or reuse.

The proposed *Nevada Historical and Environmental Geospatial Repository* seeks to address this gap through a statewide digital platform for preserving, discovering, and accessing Nevada's geospatial record. The repository is intended to serve students and faculty, public libraries, museums, K–12 educators, researchers, journalists, government agencies, nonprofit organizations, and members of the public. For rural libraries, small museums, educators, and independent researchers in particular, a shared repository would provide access to collections and technical capabilities that would be difficult to develop locally.

This is not simply a website project. It is a long-term stewardship commitment involving collections, metadata, preservation, application development, infrastructure, governance, staffing, partnerships, and sustained financial support. The technology selected for the repository will matter, but its continued success will depend just as much on whether the resulting service can be understood, operated, maintained, and migrated by future staff.

Building on UNR's Geospatial Legacy

The repository builds on several generations of geospatial leadership at the University of Nevada, Reno. Mary Ansari helped establish the Mary B. Ansari Map Library; Linda Newman developed the *Nevada in Maps Historical Digital Map Library*; and Tod Colegrove led the

redevelopment of the Nevada Geospatial Data Gateway. Together, these efforts demonstrated UNR's longstanding commitment to connecting Nevada's people with its places.

The Nevada Geospatial Data Gateway was innovative for its time and received statewide recognition. Its eventual decommissioning, however, offers an important lesson for the present project. The Gateway depended on ArcGIS Server and on specialized geospatial and web-development expertise. When funding was not available for software upgrades and dedicated technical staffing, the application could no longer be sustained. Although its underlying data survived in cloud storage, the public discovery environment did not.

A future repository should preserve this history without repeating its technical vulnerabilities. Platform selection cannot be separated from maintenance requirements, licensing obligations, staffing capacity, security updates, data portability, and the cost of replacing components as technologies change. A successful repository must be designed for its second decade—not only for its launch.

Strategic Planning Grant

The IMLS–LSTA Strategic Planning Grant represents Phase 1 of a larger, multi-phase effort. Its purpose is to evaluate lessons from earlier systems, assess statewide needs, and develop a practical roadmap for a scalable, accessible, and sustainable repository. The grant anticipates a strategic plan, partnership framework, and infrastructure needs assessment that can guide future fundraising, development, and implementation.

This planning phase is therefore appropriately concerned with questions that precede a production deployment:

- What collections and communities should the repository serve?
- Which responsibilities should remain with UNR, and which can be shared with institutional or community partners?
- How should archival masters, access derivatives, metadata, and application services relate to one another?
- Which components can be supported through established open-source communities?
- What staffing, hosting, security, and maintenance commitments will be required?
- How can UNR reduce dependency on a single vendor, product, or specialist?
- What can be demonstrated during Phase 1 without prematurely building a production system?

The work documented in this report is intended to help answer those questions. Its purpose is not to select technology in isolation. It is to connect technical decisions to the grant's broader commitments to public access, digital equity, statewide collaboration, place-based learning, and long-term stewardship.

Statement of Work

UNR retained a Geospatial Repository Specialist to provide subject-matter expertise in repository architecture and infrastructure planning. The consultant's primary deliverable is a *Technical Architecture and Sustainability Roadmap for the Nevada Historical and Environmental Geospatial Repository*.

The Statement of Work calls for an evaluation of repository platforms, web applications, hosting environments, storage systems, indexing technologies, and delivery methods. It includes consideration of open-source and proprietary options; cloud, institutional, and hybrid hosting models; metadata standards and interoperability; implementation and operating costs; staffing requirements; and long-term sustainability. It also calls for a focused prototype demonstrating representative data ingestion, metadata presentation, search, and discovery functionality.

Each component directly supports the grant's planning objectives. The platform and hosting evaluation informs the infrastructure needs assessment. Metadata and interoperability recommendations establish a foundation for statewide discovery and resource sharing. Cost and staffing analysis supports realistic fundraising and operational planning. The prototype reduces uncertainty by testing important assumptions against representative UNR resources.

The prototype should be understood as a planning artifact rather than a production repository. Its value lies in demonstrating the recommended architectural direction, exposing implementation questions, and creating reusable documentation and source code for a subsequent phase. Production deployment should follow only after UNR has confirmed governance, staffing, funding, preservation responsibilities, and service ownership.

Consultant Qualifications

Eric Larson was engaged as the Geospatial Repository Specialist because his experience directly matches the project's technical and community requirements. His work spans academic-library geospatial discovery, OpenGeoMetadata, GeoBlacklight, repository architecture, metadata lifecycle, cloud-optimized asset delivery, and the development of interoperable geospatial discovery services for the Big Ten Academic Alliance. The prototype work documented in this report demonstrates that experience in practice through a UNR-themed discovery interface, the OpenGeoMetadata API, and OpenGeoMetadata Studio.

Metadata Best Practices for Geospatial Collections

Section Summary

UNR should adopt OpenGeoMetadata Aardvark as its primary discovery metadata standard, using the BTAA extension only where additional workflow fields are needed. Metadata should be maintained as portable, version-controlled records, validated before publication, and linked to authoritative archival descriptions and cloud-optimized assets. This approach supports GeoBlacklight, the OpenGeoMetadata API, and future platforms without locking UNR into one application.

Guidance for Libraries and Cultural Heritage Institutions

For libraries and cultural heritage institutions, high-quality metadata is essential to the discovery, access, management, and reuse of collections. Metadata supports the full lifecycle of digitized and born-digital resources, including acquisition, description, publication, preservation, and long-term stewardship.

Library-led geospatial discovery systems commonly use metadata elements drawn from the Dublin Core Metadata Initiative, the Data Catalog Vocabulary, and the OpenGeoMetadata Aardvark application profile. These standards serve complementary purposes. Dublin Core provides a broadly applicable descriptive foundation; DCAT promotes interoperability among web-based data catalogs; and Aardvark adapts elements from these and other vocabularies to support geospatial discovery.

A strong metadata foundation enables essential discovery-system functions, including keyword and spatial search, faceted browsing, resource previews, downloads, record relationships, metadata harvesting, and automated citation generation. The following sections describe the standards and application profiles most commonly used by the open-source geospatial library community.

Foundational Standards and Application Profiles

Although closely related, the resources discussed below are not equivalent. Dublin Core and DCAT are general-purpose vocabularies. Aardvark and the GeoBTAA Metadata Application Profile apply selected terms and additional requirements to particular communities and systems.

Dublin Core

The [Dublin Core Metadata Initiative's Metadata Terms](#) provide one of the most widely adopted vocabularies for describing cultural heritage and information resources. Familiar elements such as title, creator, publisher, subject, description, date, language, identifier, and rights provide a common descriptive foundation across repositories and platforms.

Dublin Core remains especially valuable because its terms are broadly understood, extensible, and compatible with other metadata vocabularies. Aardvark uses many terms from the “[dcterms](#)” namespace, represented in its field names by the `dct_` prefix.

When using Dublin Core terms, institutions should apply consistent name forms, normalized dates, persistent identifiers, controlled vocabularies where appropriate, and clear rights and licensing statements. Local documentation should also distinguish frequently confused roles, particularly creator, publisher, provider, and rights holder.

DCAT — Data Catalog Vocabulary

The [Data Catalog Vocabulary, Version 3](#) is a W3C RDF vocabulary designed to support interoperability among data catalogs published on the Web. DCAT provides a standard model for describing catalogs, datasets, dataset series, distributions, and data services. Its use supports metadata aggregation, federated discovery, and exchange among otherwise independent catalog systems.

[DCAT-US 3.0](#) is the United States application profile of DCAT 3 and serves as the federal metadata standard for documenting datasets, APIs, and data services. It adds requirements and implementation guidance appropriate to federal data-management and reporting needs.

For cultural heritage institutions, DCAT is particularly useful when geospatial collections must be exposed to broader research-data catalogs or open-data infrastructure. It should not, however, be treated as a complete technical or preservation-metadata standard.

OpenGeoMetadata

[OpenGeoMetadata](#) is both a discovery-metadata framework for geospatial resources and an open platform for sharing metadata files. Its community develops shared conventions, documentation, validation resources, conversion tools, and repositories that enable participating institutions to exchange geospatial discovery metadata.

The community's collaborative approach reduces local variation and makes it easier for institutions to aggregate metadata from multiple repositories.

OGM Aardvark Metadata Application Profile

The [OpenGeoMetadata Aardvark schema](#) is a discovery-focused metadata application profile for geospatial resources. Developed collaboratively by metadata specialists and librarians from twelve institutions, Aardvark was designed to improve interoperability while supporting a wider range of resources than the earlier GeoBlacklight 1.0 schema.

Aardvark combines elements from Dublin Core, DCAT, schema.org, the Portland Common Data Model, and other vocabularies with application-specific GeoBlacklight elements. The result is a largely flat, JSON-based profile optimized for indexing, search, discovery, and metadata exchange.

[GeoBlacklight uses Aardvark](#) as its current metadata profile. Aardvark fields support text and spatial search, facets, resource-type icons, access controls, downloads, service previews, related-resource displays, and citation generation.

Aardvark is intentionally lightweight and discovery-oriented. It is not intended to replace detailed technical metadata, preservation metadata, data dictionaries, or records of a dataset's processing history. These materials should be retained as supplemental metadata and linked from the Aardvark record when possible.

GeoBTAA Metadata Application Profile

The Big Ten Academic Alliance has formalized its implementation of Aardvark through the [GeoBTAA Metadata Application Profile](#). The BTAA Geoportal uses the standard Aardvark elements while applying additional guidance to selected field values.

The profile also defines custom fields using the **b1g** namespace. These fields support local acquisition, administration, lifecycle tracking, processing, and archival workflows that fall outside Aardvark's discovery-focused scope.

Detailed supplemental records—including ISO 19139, FGDC, MODS, and other technical or descriptive metadata—may be retained separately and linked from the discovery record.

Aardvark Validation

Every Aardvark record should be validated before it is indexed, published, or shared. OpenGeoMetadata publishes [JSON Schema resources](#) that can be used to check record structure programmatically.

Schema validation should confirm:

- Valid JSON syntax and structure
- Presence of mandatory fields

- Correct field names and data types
- Appropriate handling of single and multivalued fields
- Compliance with strict controlled vocabularies
- Valid date, Boolean, geometry, and bounding-box values
- Correctly structured reference links
- Unique and stable record identifiers

Validation should be integrated into batch-processing and continuous-integration workflows whenever possible. Records that fail mandatory checks should not be published until the errors have been corrected.

Aardvark Quality Evaluation

A record can be technically valid and still provide poor discovery or access. Metadata quality should therefore be evaluated separately from schema conformance.

A local quality scorecard may assess:

- **Conformance:** Are mandatory fields present and correctly formatted?
- **Discoverability:** Are the title, description, subjects, and resource classifications meaningful?
- **Spatial and temporal description:** Can users determine where and when the resource applies?
- **Access and operability:** Do download, service, preview, and supplemental-metadata links function?
- **Rights and provenance:** Are access conditions, licenses, providers, creators, and publishers clearly identified?
- **Maintenance:** Does the record use persistent identifiers, document modification dates, and identify related or superseded resources?
- **Consistency:** Are names, dates, places, formats, and controlled values applied consistently across the collection?

The scorecard should distinguish among errors that prevent successful ingestion, warnings that impair system functionality, and recommendations that improve discovery or citation quality.

Metadata Crosswalks

Metadata crosswalks define how elements and values are transformed from one schema into another. They should be treated as documented transformations rather than assumptions of one-to-one equivalence.

Each crosswalk should identify:

- Direct, conditional, normalized, and derived mappings

- Source fields that cannot be represented in the target schema
- Target fields that require enrichment beyond the source record
- Rules for repeated values and compound fields
- Controlled-vocabulary transformations
- Date and spatial normalization procedures
- Provenance and modification-date handling
- Examples and automated tests for representative records

MARC to Aardvark

A MARC-to-Aardvark crosswalk can derive titles, names, publication information, dates, subjects, identifiers, and electronic-access links from bibliographic records. Additional enrichment will often be required for bounding boxes, geometry, resource class, resource type, provider, access rights, and actionable download or service links.

Special attention should be given to cartographic mathematical data, geographic subject headings, edition statements, related works, and distinctions among creators, publishers, and custodial institutions.

MODS to Aardvark

MODS provides structured descriptive metadata that often maps effectively to Aardvark titles, names, origin information, subjects, identifiers, locations, and related resources. Because MODS can preserve greater bibliographic detail than Aardvark, the original MODS record should generally be retained and linked as supplemental metadata.

Dublin Core to Aardvark

Many Dublin Core terms map directly to Aardvark fields. A complete transformation must nevertheless supply Aardvark-specific administrative, spatial, resource-classification, and reference fields required for geospatial discovery and GeoBlacklight functionality.

Metadata Concerns

Resource Metadata

Resource metadata describes the intellectual content and discovery characteristics of an individual resource. It should answer the fundamental questions users bring to a catalog: What is this resource? Who created or published it? Where and when does it apply? How may it be accessed and reused?

Resource metadata should remain closely associated with the resource and should be updated when its content, location, access conditions, or relationships change.

Archival Metadata

Archival metadata supports preservation, authenticity, provenance, and long-term management. It may include file-level technical characteristics, checksums, preservation events, transfer history, processing documentation, package relationships, retention decisions, and evidence of format migration.

Aardvark should function as the discovery layer rather than the sole archival record. Preservation metadata should be maintained in an appropriate repository or preservation system and linked to the discovery record when useful.

AI-Enrichment Metadata

AI-assisted enrichment can improve titles, descriptions, subjects, geographic names, dates, and other discovery fields. AI-generated metadata should be treated as derived information with documented provenance—not as an undocumented replacement for source or human-authored metadata.

Institutions should record, where practicable:

- Which values were generated or modified using AI
- The model, service, and version used
- The date and method of generation
- The source materials supplied to the system
- Confidence scores or uncertainty indicators
- The validation status and reviewing staff member
- Subsequent human corrections or approvals

Original values should be retained, and AI-generated suggestions should remain distinguishable from approved metadata. Enrichment workflows should also address privacy, copyright, cultural sensitivity, bias, and the risk of unsupported geographic or historical assertions.

Recommendation for the University of Nevada, Reno

UNR should preserve authoritative source metadata, generate Aardvark records as a documented discovery representation, retain richer supplemental and preservation metadata, validate every transformed record, and evaluate descriptive quality separately from technical validity. Before production migration, UNR should publish and maintain a local Aardvark application profile defining required, recommended, and optional fields; controlled vocabularies; identifier, rights, and publication-state policies; validation thresholds; review procedures; and representative records.

Crosswalks, enrichment rules, controlled vocabularies, and validation criteria should be versioned and maintained as institutional documentation. This approach supports dependable

local operations while improving interoperability across GeoBlacklight, OpenGeoMetadata, GeoBTAA, DCAT, and emerging OGC catalog environments.

Cloud-Optimized Assets

Section Summary

UNR should preserve original files while delivering standardized cloud-optimized derivatives: COGs for raster data, PMTiles for vector data, and IIIF for scanned maps. These assets can be validated, stored inexpensively in S3-compatible object storage, and previewed directly in modern browsers—reducing the cost, complexity, and security exposure of operating GeoServer.

For libraries and cultural heritage institutions, digital assets must serve several purposes at once. They must preserve the original object, support long-term stewardship, provide reliable downloads, and enable responsive previews within a discovery environment. These responsibilities should not be assigned to a single file—or made dependent upon a single application server.

Cloud-optimized formats offer a more sustainable approach. Instead of asking a map server to open, transform, and render every resource on demand, these formats organize data so that a browser or desktop application can retrieve only the portions it needs. The resulting files can be stored on Amazon S3 or another S3-compatible platform and delivered through ordinary HTTP requests.

This model separates preservation from presentation. Original files remain intact as archival objects, while standardized derivatives support discovery, visualization, and reuse. It also reduces the number of public-facing services that must be maintained, patched, and monitored.

A solid asset foundation is every bit as important as a solid metadata foundation. Let's review the formats and practices most relevant to the University of Nevada, Reno's future geoportal.

Original Archival Files

Original files should remain the authoritative preservation copies of UNR's digital geospatial collections. They should be retained in the format received or created whenever that format can be preserved safely.

These files may include:

- Uncompressed or losslessly compressed TIFF images
- GeoTIFFs in their original coordinate reference systems
- Shapefiles and their associated component files
- File geodatabases

- GeoPackages
- Scanned maps and aerial photographs
- Project documentation, metadata, and ancillary files

Archival originals should not be altered simply to improve web performance. They should be stored in managed preservation storage with fixity information, provenance, access controls, and documented relationships to any derivatives created from them.

The original file and the preferred access file may sometimes be identical, but this should be a deliberate determination rather than an assumption. A valid GeoTIFF, for example, is not necessarily a valid Cloud Optimized GeoTIFF. Likewise, a preservation-quality TIFF may be much larger than the derivative needed for routine browser display.

Cloud-Optimized Derivatives

Cloud-optimized derivatives should provide the principal access and preview layer for the future geoportal. These files should be reproducible from the archival original and treated as managed service derivatives rather than preservation masters.

The recommended formats are:

- IIIF Level 0 for scanned maps and other non-georeferenced imagery
- Cloud Optimized GeoTIFF for georeferenced raster data
- PMTiles for tiled vector previews and, where appropriate, raster tile pyramids
- GeoJSON for small and moderately sized vector resources
- FlatGeobuf for cloud-accessible feature data
- GeoParquet for analytical and data-science use
- GeoPackage or another familiar GIS format for desktop download
- JPEG, PNG, or WebP for thumbnails and static map images

No single derivative will satisfy every use case. A resource may therefore have several related representations: an archival original, a downloadable access copy, an interactive preview asset, a thumbnail, and a static map. The metadata record should identify the purpose and format of each representation clearly.

IIIF Level 0

The [International Image Interoperability Framework](#) remains an excellent option for scanned maps, aerial photographs, and other image-based collections. IIIF supports interoperable image viewers while providing a consistent way to describe image dimensions, tile sizes, scale factors, and available representations.

A Level 0 IIIF implementation is particularly attractive for UNR because it can be delivered entirely from static object storage. Image tiles and an `info.json` document are generated in

advance and published through an S3-compatible endpoint. The browser then requests the appropriate tiles without relying upon a continuously running image server.

Level 0 does not support the complete set of dynamic image transformations available from a full IIIF Image API server. For discovery previews, however, pre-generated tiles are frequently sufficient. This offers a practical balance between interoperability, performance, and operational simplicity.

UNR already has several useful Level 0 prototypes:

- [Ely, Nev., September 1923, sheet 6](#)
- [Township No. 13 North, Range No. 18 East](#)
- [Bullfrog quadrangle, Nevada-California](#)
- [Reno sheet, Nevada](#)
- [Topographical map of the Reese River mines](#)

These examples demonstrate that UNR can generate interoperable tiled-image derivatives without operating a full IIIF image server. The same approach can be incorporated into a repeatable accession and processing workflow.

Cloud Optimized GeoTIFFs

A [Cloud Optimized GeoTIFF](#) is a GeoTIFF organized internally for efficient access over HTTP. Image data is tiled, lower-resolution overviews are embedded within the file, and the most frequently needed structural information is placed where a client can retrieve it efficiently.

When the storage platform supports HTTP byte-range requests, a browser or GIS application can read a small portion of a COG without downloading the entire raster. This makes COGs well suited to object storage, content-delivery networks, desktop GIS software, analytical environments, and browser-based map viewers.

UNR should adopt COG as the preferred access derivative for georeferenced raster resources. Each COG should include:

- A documented coordinate reference system
- Internal tiling
- Appropriate internal overviews
- A defined NoData value or internal mask when necessary
- Compression appropriate to the source material
- Embedded or associated geospatial metadata
- A persistent relationship to the archival original

Lossless compression should remain the default for maps, scientific rasters, and other materials where pixel values or fine linework must be preserved. Lossy compression may be appropriate

for certain continuous-tone photographic images, but that choice should be documented and tested visually.

UNR's existing COG prototypes include:

- [Boulder City, 1952](#)
- [Quad 0101 UTM Zone 11 Georeference](#)
- [o35114a5 GeoTIFF raster dataset](#)

These files have the expected COG structure, including internal tiling and overviews, and are delivered from storage that supports HTTP range requests. They provide a sound starting point for UNR's raster workflow.

Native-Resolution and Preview COGs

A COG's coordinate reference system requires special consideration. Desktop GIS and analytical tools can generally read COGs in their native projections, but browser-based viewers do not necessarily support every coordinate system. Some GeoBlacklight and MapLibre integrations currently expect Web Mercator or geographic coordinates and cannot reproject arbitrary rasters in the browser.

UNR should therefore distinguish between an analysis COG and a preview COG when necessary:

- The analysis COG should preserve the raster's preferred or native coordinate reference system.
- A preview COG may be reprojected to Web Mercator for predictable display in the geoportal.
- The archival original should remain unchanged.
- Metadata should identify each file's purpose and coordinate reference system.

This will sometimes result in two COG derivatives, but it avoids weakening the analytical value of the resource simply to accommodate a browser viewer. If UNR elects to operate a lightweight raster service such as TiTiler, dynamic preview tiles could instead be generated from the native COG. That approach reduces duplicate storage but reintroduces an application service that must be operated and maintained.

PMTiles

[PMTiles](#) packages a tiled map pyramid into a single file that can be read using HTTP byte-range requests. It can therefore be placed in object storage and accessed directly by a compatible browser viewer without a conventional tile server.

PMTiles is especially useful for vector previews. Large feature collections can be generalized and divided across zoom levels, giving users a responsive overview without requiring the browser to download the complete source dataset.

UNR's current PMTiles examples include:

- [Ortho 1-1 1n NV510, 2010: Polygon data](#)
- [NAIP Nevada 2010 1-meter 4-band image tile footprints](#)
- [NDOT Georeferenced TIFF Index Polygons](#)

These files retain feature attributes, demonstrating that a PMTiles preview does not need to be limited to geometry alone.

PMTiles and Attributes

A curated subset of attributes should be included when those attributes support useful interactions such as identification, labeling, filtering, or linking to related resources. Examples might include a sheet number, image date, place name, identifier, or source URL.

The complete source attribute table should not be included automatically. Large, redundant, sensitive, or difficult-to-interpret fields can increase file size and expose information that was never intended for public display. Attribute selection should therefore be part of the derivative profile for each collection.

PMTiles should be treated as a display derivative rather than the authoritative data download. Vector tiles may simplify geometries, omit features at lower zoom levels, normalize data types, or contain only selected attributes. Users needing the complete dataset should be directed to a GeoPackage, FlatGeobuf, GeoParquet, or original source package.

GeoJSON

GeoJSON remains one of the most widely understood formats for exchanging vector data on the Web. It is human-readable, well supported by JavaScript mapping libraries, and convenient for small collections.

UNR should use GeoJSON for:

- Small feature collections
- Simple footprints and index layers
- Search-result overlays
- Bounding geometries
- Lightweight downloads
- Diagnostic and interoperability workflows

Attributes may be retained when they are appropriate for public distribution. Because an ordinary GeoJSON file is generally downloaded and parsed as a whole, however, performance

declines as file size and feature complexity increase. It should not become the default preview format for every vector dataset.

Large or complex collections should use PMTiles for display and a more scalable format for complete feature delivery.

FlatGeobuf

[FlatGeobuf](#) is a compact binary vector format designed to support efficient spatial filtering, including access over HTTP. Unlike PMTiles, it can retain complete feature geometries and attribute tables without converting them into a display-oriented tile pyramid.

FlatGeobuf is a strong candidate for cloud-based vector distribution. It can support workflows that need the complete features intersecting a particular area without requiring the user to download an entire large dataset.

For UNR, FlatGeobuf can serve as:

- A cloud-accessible vector derivative
- A complete feature download
- An intermediate format in derivative-generation workflows
- A complement to PMTiles previews

Client support is not yet as universal as support for GeoJSON or GeoPackage, so FlatGeobuf should be offered as part of a portfolio of access formats rather than as the only vector download.

GeoParquet

[GeoParquet](#) extends the Apache Parquet format with standardized geospatial information. It is designed primarily for analysis rather than interactive cartographic display.

GeoParquet is particularly useful for large vector collections that will be examined in Python, R, DuckDB, data warehouses, or other column-oriented analytical environments. It provides efficient compression and allows analytical tools to read only the columns needed for a particular operation.

UNR should consider GeoParquet for large, attribute-rich datasets and research collections likely to support computational use. It should not replace PMTiles as the browser-preview format or GeoPackage as the most familiar desktop-GIS download. Its value lies in extending the collection to emerging data-science workflows.

Thumbnails and Static Maps

Interactive previews are useful, but they should not be the only visual representation of a resource. Every publicly discoverable spatial record should have a lightweight thumbnail, and significant resources should also have a larger static map or representative image.

Thumbnails support search results, social previews, accessibility workflows, and situations in which the interactive viewer cannot be loaded. Static maps provide a dependable fallback when a browser lacks support for a particular cloud-optimized format or when a derivative temporarily fails validation.

Recommended outputs include:

- A small search-result thumbnail
- A larger item-page preview
- A static map showing the resource within its geographic context
- A representative image for non-spatial or tabular resources
- A placeholder generated from the resource's geometry when no other image is available

These images should be created during derivative processing rather than generated dynamically for every request.

Optimization and Compression

Optimization should be tailored to the characteristics of the resource. The goal is not simply to make each file as small as possible, but to balance visual quality, analytical integrity, transfer size, and decoding performance.

Raster processing should consider:

- Internal tile dimensions
- Overview levels
- Lossless or lossy compression
- Color depth
- Alpha channels and masks
- NoData values
- Photometric interpretation
- Browser-compatible preview projections

Vector processing should consider:

- Geometry repair
- Coordinate precision
- Simplification by zoom level
- Attribute selection

- Feature density
- Minimum and maximum display zooms
- Duplicate or empty geometries

Processing decisions should be documented in reusable derivative profiles. This makes the workflow repeatable and allows derivatives to be regenerated when standards, software, or access requirements change.

Asset Validation

A file's existence in object storage does not demonstrate that it is valid or useful. Validation must test the asset itself, its delivery environment, and its presentation in the geoportal.

Structural Validation

Automated checks should confirm that:

- COGs satisfy COG layout requirements and contain usable overviews.
- IIIF [info.json](#) documents are valid and reference available tiles.
- PMTiles files contain readable headers, metadata, zoom levels, and tile directories.
- GeoJSON and FlatGeobuf files contain valid geometries.
- GeoParquet files include the expected geospatial metadata.
- Files have expected sizes, checksums, media types, and extensions.

The [GDAL COG validator](#) can form part of the raster validation workflow.

Geospatial Validation

The workflow should also inspect:

- Coordinate reference systems
- Spatial extents
- Pixel dimensions and resolutions
- Feature and record counts
- Geometry types
- NoData and mask behavior
- Raster statistics
- Attribute names and data types

Unexpected changes between the source and derivative should either fail validation or be recorded as intentional transformations.

HTTP Validation

Cloud-optimized delivery depends upon correct storage configuration. Automated tests should confirm:

- Successful **HEAD** and **GET** requests
- HTTP byte-range support
- Correct **Content-Type** values
- Cross-origin resource sharing headers
- Appropriate cache controls
- Stable public or signed URLs
- A **206 Partial Content** response for byte-range requests
- Continued availability of each URL referenced by the metadata

A technically valid COG or PMTiles file will not function as intended if the storage platform prevents range requests or blocks the geoportal through an incorrect cross-origin policy.

Visual Validation

Each preview should be opened in the same viewer used by the production geoportal. Visual checks should look for:

- Incorrect extents or projections
- Blank previews
- Upside-down or mirrored imagery
- Missing tiles
- Unexpected transparency
- Poor color or contrast
- Excessive vector simplification
- Missing labels or popup attributes
- Unusable performance at broad or detailed zoom levels

Representative samples from each collection should receive human review. Automated screenshots may also be compared against approved reference images to identify regressions.

Metadata and Asset Validation

Metadata and asset validation should be joined together. An Aardvark record should not pass final quality review merely because its descriptive fields are complete. Its referenced thumbnail, preview, download, IIIF, COG, or PMTiles assets must also resolve and behave as expected.

The [OpenGeoMetadata reference URI registry](#) should be used consistently so that GeoBlacklight and other applications can distinguish among downloads, services, thumbnails, and cloud-optimized assets.

Prior Art: Stanford, Princeton, and Harvard

Stanford University Libraries is not alone in reconsidering GeoServer's role within a production GeoBlacklight environment. Princeton and Harvard have also taken significant steps to reduce or remove their dependence on GeoServer for resource previews and data delivery. Together, these institutions demonstrate a broader movement away from maintaining a general-purpose map server when static, cloud-optimized assets can satisfy the majority of discovery and access requirements.

Stanford's recent work provides a particularly useful operational case study. Following security concerns that required its GeoServer instance to be taken offline, Stanford expanded its use of Cloud Optimized GeoTIFFs, PMTiles, IIIF, and other static assets delivered from object storage. Its approach reduces the number of public-facing services that must be patched and monitored while allowing browser viewers to retrieve only the data required for the current display. [Stanford's Geo4LibCamp presentation](#) provides a practical model for this transition.

Princeton has documented an especially thorough migration. Its repository infrastructure replaced raster delivery through GeoServer with TiTiler, generated COG and PMTiles derivatives in Figgy, added COG and PMTiles support to its GeoBlacklight application, and migrated vector resources to PMTiles. Princeton subsequently closed its [GeoServer decommissioning work](#) and removed the GeoServer deployment role from its infrastructure configuration.

Harvard has likewise taken significant steps to reduce GeoServer's role in its production GeoBlacklight environment. Its transition is instructive because it demonstrates that modernization does not need to occur as a single clean break. Legacy records and service references may coexist with newer delivery mechanisms while collections, metadata, and applications are migrated incrementally.

The discussions at Geo4LibCamp 2026 reinforce this direction. Cloud-optimized formats are no longer peripheral experiments within the library geospatial community. They are becoming practical infrastructure components for preview, distribution, preservation access, and cross-repository interoperability.

Recommendation for the University of Nevada, Reno

UNR should adopt an object-storage-first architecture for the future geoportal. GeoServer should not be the default mechanism for previewing or distributing static collection resources.

The recommended asset model has three layers:

1. **Preservation assets:** Retain original files, associated documentation, checksums, and provenance in managed archival storage.

2. **Access assets:** Generate standardized derivatives such as native-resolution COGs, FlatGeobuf, GeoParquet, GeoPackage, and packaged original downloads.
3. **Presentation assets:** Generate IIIF Level 0 tiles, Web Mercator preview COGs where needed, PMTiles, thumbnails, and static maps for direct use by the geoportal.

Metadata should connect these layers through persistent, typed references. Derivatives should be reproducible, versioned, and validated automatically before their URLs are added to the public record.

UNR should express these requirements as versioned derivative profiles that record the source format, intended use, coordinate reference system, compression, tiling and overview settings, attribute-selection rules, naming conventions, validation commands, and regeneration procedure for each supported output.

GeoServer—or a smaller purpose-specific service such as TiTiler—should be retained only where requirements genuinely call for dynamic processing. These cases may include:

- On-demand reprojection
- Dynamic styling
- OGC WMS or WFS access
- Server-side feature queries
- Transactional editing
- Authenticated or restricted resources
- Raster calculations or transformations
- Resources that cannot be represented adequately through static derivatives

This is not an argument that GeoServer has no continuing value. It is a recommendation that UNR not make every static resource dependent upon it.

Implementation should proceed incrementally. UNR can begin with the IIIF, COG, and PMTiles prototypes already generated through OGM Metadata Studio; formalize derivative profiles and validation rules; add the resulting asset references to Aardvark metadata; and confirm that the future GeoBlacklight interface can display them without a map server. Legacy records can then be migrated collection by collection.

This approach places UNR in alignment with the architectural direction being pursued by Stanford, Princeton, Harvard, and the broader GeoBlacklight community. It also turns GeoServer retirement from an emergency response into a planned infrastructure objective: generate replacement assets first, measure the remaining service dependencies, and retire server components as their functions are replaced.

Geoportal Options

Section Summary

UNR should pursue an open-source geoportal rather than commit immediately to a commercial ArcGIS platform. Begin with a pilot using the community-sponsored OpenGeoMetadata API and a lightweight customizable frontend; retain GeoBlacklight as the established fallback if the newer approach does not meet UNR's requirements. Both options support Aardvark metadata and cloud-optimized assets, preserving UNR's investment and avoiding vendor lock-in.

A geoportal is the public expression of the metadata and digital assets described in the preceding sections. It brings together search, spatial discovery, previews, downloads, citations, and contextual information within a coherent interface. The choice of geoportal software will therefore shape not only what users see, but also how UNR manages metadata, generates derivatives, integrates with other systems, and sustains the service over time.

No platform should become the sole repository for UNR's descriptive metadata or digital assets. Canonical Aardvark records, archival originals, and cloud-optimized derivatives should remain portable and independently managed. The geoportal should index, interpret, and present those resources without becoming their only viable home.

With that foundation in place, UNR has several credible commercial and open-source options.

Commercial Options

The ArcGIS Ecosystem

[ArcGIS](#) is the dominant commercial platform for creating, managing, analyzing, and publishing geospatial information. It offers a broad collection of desktop, server, cloud, mobile, and web applications built around a shared content and identity model.

For a geoportal project, the relevant choices are not simply “ArcGIS” or “not ArcGIS.” UNR would need to choose among ArcGIS Online and ArcGIS Hub as vendor-hosted services, or ArcGIS Enterprise and Enterprise Sites as institutionally operated software.

ArcGIS Online and ArcGIS Hub

[ArcGIS Online](#) is Esri's software-as-a-service environment. It manages users, groups, hosted layers, maps, applications, and other ArcGIS content without requiring the institution to operate the underlying platform.

[ArcGIS Hub](#) builds public-facing websites and searchable catalogs from content managed through ArcGIS Online. Hub Basic is included with an ArcGIS Online organizational subscription and supports sites, pages, content catalogs, groups, and public sharing.

This is an attractive option when an institution already uses ArcGIS Online extensively. Campus GIS staff may already understand its publishing workflows, identity model, access controls, and service architecture. A Hub site can be assembled relatively quickly, and Esri assumes responsibility for much of the platform's availability and software maintenance.

For UNR, ArcGIS Hub would work best if:

- ArcGIS Online is already the preferred campus publishing environment.
- Most contributing units create ArcGIS items or hosted services.
- Vendor support is valued more highly than application-level control.
- The geoportal is expected to include operational campus GIS data alongside library collections.
- The institution wants to minimize responsibility for maintaining the portal software itself.

The principal limitation is that ArcGIS Hub organizes its catalog around the ArcGIS item model. Aardvark metadata, IIIF resources, PMTiles, and independently hosted COGs can be linked or represented, but they are not the platform's native center of gravity. Preserving UNR's richer library metadata may require crosswalks, custom fields, additional ArcGIS items, or a parallel metadata repository.

Interactive support for cloud-optimized assets should also be tested carefully. ArcGIS applications work particularly well with ArcGIS feature, map, imagery, and tile services. Direct presentation of IIIF, PMTiles, or other emerging formats may require custom applications even when the underlying files remain available through Hub.

ArcGIS Hub should therefore be understood as a strong ArcGIS-native publishing solution, not as a direct substitute for the OpenGeoMetadata and GeoBlacklight ecosystem.

ArcGIS Hub Premium

ArcGIS Hub Premium expands Hub Basic with community accounts and engagement features such as initiatives, projects, discussions, events, and collaborative participation. It also includes a separate ArcGIS Online community organization through which external participants can create accounts and contribute content.

These capabilities are valuable for civic-engagement programs, participatory mapping, volunteer projects, and long-running initiatives involving people outside the institution. They are not automatically necessary for a library discovery portal.

UNR should consider Hub Premium only if the geoportal is expected to support active external participation. If the primary requirement is to search, preview, and download curated collections, Hub Basic may provide the relevant site and catalog functions without the additional organizational and licensing complexity.

ArcGIS Enterprise Sites

[ArcGIS Enterprise Sites](#) provides a Hub-like site-building experience within a self-hosted ArcGIS Enterprise deployment. It may be appropriate when institutional requirements prevent content from being hosted in ArcGIS Online or when UNR already operates ArcGIS Enterprise as a supported campus service.

Enterprise Sites provides greater control over infrastructure, authentication, storage, and upgrade timing. That control also returns responsibility for operating Portal for ArcGIS, ArcGIS Server, supporting data stores, certificates, backups, monitoring, and version upgrades to the institution.

For a library seeking to reduce its dependence on public-facing application servers, a new ArcGIS Enterprise deployment would move in the opposite direction. It should be considered only when UNR already has an Enterprise environment and a clearly identified unit willing to own the service. ArcGIS Enterprise is too substantial an infrastructure commitment to deploy solely for a digital-collections geoportal.

Minnesota Geospatial Commons

The [Minnesota Geospatial Commons](#) is a useful example of ArcGIS Hub as a statewide discovery platform. Minnesota operated its previous Commons using CKAN and migrated to a modernized ArcGIS Hub implementation in 2026.

The new Commons demonstrates Hub's strengths for a distributed publishing community: multiple organizations, standardized ArcGIS services, public data downloads, resource categories, and publisher-oriented workflows. It also illustrates the appeal of transferring more platform maintenance to a commercial cloud service.

Minnesota's use case is not identical to UNR's. A statewide government data clearinghouse primarily distributes current administrative and operational data, while a library geoportal must also describe historical maps, scanned documents, archival collections, and resources held in non-ArcGIS repositories. The Commons is therefore useful prior art, but its architecture should not be adopted without testing UNR's cultural-heritage metadata and cloud-optimized assets.

Open-Source Options

GeoBlacklight

Project: geoblacklight.org

Example: [Stanford EarthWorks](#)

GeoBlacklight is the most widely adopted open-source geospatial discovery application in academic libraries. It extends the Blacklight discovery framework with spatial search, map-based browsing, geospatial facets, resource previews, downloads, and geospatially aware record pages.

Its greatest advantage for UNR is its direct relationship to the metadata and asset practices already recommended in this report. GeoBlacklight treats OpenGeoMetadata Aardvark as a first-class metadata schema and participates in the same repository-sharing community described in the metadata section. Current releases can present IIIF images, COGs, PMTiles, and conventional web services, allowing GeoServer to remain optional.

GeoBlacklight is a mature and actively maintained project. Its current application stack is based on Ruby on Rails, Blacklight, Apache Solr, and associated web-application infrastructure. This provides substantial flexibility, but it also requires developers and system administrators who are comfortable operating and upgrading Rails applications.

GeoBlacklight is a strong option when:

- Aardvark will be UNR's canonical discovery schema.
- Academic-library interoperability is a central requirement.
- UNR wants to participate in OpenGeoMetadata sharing.
- The institution can sustain a Rails and Solr application.
- Local interface customization is important.
- Cloud-optimized assets will be delivered directly from object storage.

Its principal challenge is operational. GeoBlacklight is an application framework rather than a turnkey hosted service. UNR must establish responsibility for dependency upgrades, security patches, indexing, deployment, monitoring, accessibility testing, and local customizations. Organizations without an established Rails support model may find that this responsibility becomes concentrated in one developer or project position.

A GeoBlacklight implementation no longer requires the traditional GeoBlacklight-plus-GeoServer architecture. UNR could operate GeoBlacklight with Aardvark metadata and S3-compatible storage while using IIIF, COG, and PMTiles for most previews. GeoServer could then be limited to exceptional resources or omitted entirely.

BTAA Geospatial API and OpenGeoMetadata API

BTAA Geospatial API: <https://github.com/geobtaa/api>

OpenGeoMetadata API: <https://github.com/OpenGeoMetadata/ogm-api>

OpenGeoMetadata Discovery: <https://github.com/OpenGeoMetadata/ogm-discovery>

The BTAA Geospatial API represents a newer, API-first approach to geospatial discovery. Instead of binding the search index, business logic, and user interface into one application, it exposes normalized geospatial metadata and discovery functions through a documented web API.

The BTAA developed this architecture to make its collections useful beyond the classic Geoportal website. The API supports resource retrieval, search, facets, map summaries, citations, downloads, metadata representations, thumbnails, static maps, and asset distributions. It also provides an OGC API-Records-compatible interface and Model Context Protocol support.

This allows the same collection to be used through several interfaces:

- A public web geoportal
- A reusable institutional frontend
- QGIS
- A command-line client
- MCP-compatible research and AI tools
- Institutional websites and digital-library applications
- Future interfaces that do not yet exist

The associated **ogm-discovery** project is a configurable React frontend that can be branded for an individual institution and hosted as a largely static web application. It communicates with the API rather than maintaining its own independent catalog index.

This architecture is sometimes described as a Backend for Frontend pattern. More precisely, it is a headless or API-first architecture in which the discovery backend is separated from its consumers. A dedicated Backend for Frontend layer may still be added when a deployment needs server-side authentication, private credentials, response aggregation, or additional control over browser traffic. It is not required for a public static frontend using a public, rate-limited API.

For UNR, the API-first model offers several significant advantages:

- Aardvark and the BTAA Aardvark extension remain central to the data model.
- The web interface can evolve without replacing the discovery API.
- Metadata can be reused outside the portal.
- Cloud-optimized assets can be described as distributions and accessed directly.
- A lightweight frontend can be hosted separately from the application backend.

- Search and resource services can support QGIS, scripts, research tools, and AI clients.
- Other institutions can reuse the frontend without maintaining identical application stacks.

The tradeoff is maturity. GeoBlacklight has a longer production history and a larger body of institutional experience. The BTAA/OpenGeoMetadata API architecture is emerging and will require stable specifications, versioning commitments, deployment documentation, governance, and a sustainable contributor community. An early adopter must be prepared to participate in that development rather than consume a finished turnkey product.

The API is also a discovery service, not a preservation system or metadata-authoring application. UNR would still need documented workflows for creating, reviewing, storing, and updating its canonical metadata and assets.

CKAN

Project: ckan.org

Source code: github.com/ckan/ckan

CKAN is a mature, Python-based, open-source data management system maintained within the Open Knowledge Foundation ecosystem. It is used by national, regional, and institutional open-data portals around the world.

CKAN provides dataset records, organizations, groups, resources, permissions, APIs, harvesting, and configurable metadata fields. Its extension system can add spatial search, metadata harvesting, map previews, DCAT support, and other specialized capabilities.

The [CKAN spatial extension](#) adds geographic indexing and bounding-box queries, while related extensions provide GeoJSON and other map previews. This makes CKAN a credible platform for a broad institutional data repository containing geospatial and non-geospatial resources.

CKAN is strongest when:

- The institution needs a general-purpose open-data portal.
- Many departments will publish their own datasets.
- Dataset management and publisher workflows are as important as discovery.
- Python is better supported locally than Ruby.
- The collection includes substantial non-geospatial data.
- The institution is prepared to select and maintain a suite of CKAN extensions.

CKAN is less naturally aligned with academic geospatial collections than GeoBlacklight. Aardvark would require a custom schema or crosswalk, and advanced support for IIIF, COG, PMTiles, archival relationships, and library-style citations would need to be developed or integrated. Its geospatial functionality is supplied through extensions rather than forming the core of the application.

Minnesota's former Geospatial Commons demonstrated that CKAN can support a significant geospatial clearinghouse. Minnesota's subsequent migration to ArcGIS Hub should not be interpreted as a general failure of CKAN, but it does illustrate the long-term effort required to maintain a customized open-source portal and its extensions.

CKAN should be considered if UNR is evaluating a university-wide open-data platform. It is less compelling as a dedicated replacement for a library-centered GeoBlacklight service.

Common Geoportal Technology Stacks

The portal product and the asset-delivery architecture should be considered together. Several combinations are possible.

GeoBlacklight, GeoServer, and Object Storage

This is the traditional academic geoportal stack. GeoBlacklight provides discovery, GeoServer renders or exports geospatial data, and object storage holds downloads and other files.

It offers broad support for WMS, WFS, server-side styling, reprojection, and generated downloads. It also carries the largest maintenance footprint. Both GeoBlacklight and GeoServer must be secured, upgraded, monitored, and coordinated with the metadata.

UNR should not select this as its default architecture. It may remain appropriate when dynamic OGC services are an explicit requirement.

GeoBlacklight and Object Storage

In this model, GeoBlacklight handles discovery while IIIF, COG, PMTiles, GeoJSON, FlatGeobuf, and other derivatives are delivered directly from S3-compatible storage.

This is a strong modernization path for institutions that want GeoBlacklight's mature library interface without making every preview dependent upon GeoServer. It preserves compatibility with Aardvark and OpenGeoMetadata while reducing the number of public-facing services.

OpenGeoMetadata API and Object Storage

In the API-first model, the BTAA or OpenGeoMetadata API provides search and resource information, a frontend such as [ogm-discovery](#) provides the public interface, and cloud-optimized assets are delivered from object storage.

This creates the cleanest separation among metadata, discovery services, presentation, and asset delivery. It also provides the greatest opportunity for reuse through QGIS, command-line tools, MCP clients, and future applications.

Its success depends upon sustained API governance and production support.

ArcGIS Hub and ArcGIS-Managed Services

In an ArcGIS-centered model, ArcGIS Online or Enterprise manages catalog items, identities, hosted layers, and applications. Hub or Enterprise Sites supplies the public discovery interface.

This can minimize custom development and align the geoportal with campus GIS operations. It may also increase dependence upon Esri licensing, user types, credits, service formats, and metadata conventions. UNR should preserve canonical metadata and archival assets outside the ArcGIS item model even if Hub is selected.

The Role of GeoServer

GeoServer is not itself a geoportal. It is a geospatial service platform that can publish data through interfaces such as WMS, WFS, WMTS, and OGC API services.

Historically, GeoServer supplied the preview and download functions that discovery applications could not perform themselves. Cloud-optimized formats now replace many—but not all—of those responsibilities.

GeoServer remains appropriate for:

- Dynamic server-side styling
- On-demand reprojection
- Feature-level spatial and attribute queries
- WMS or WFS requirements
- Transactional editing
- Protected services
- Dynamic rendering from databases
- Workflows that cannot be represented by static derivatives

It should not be operated merely to create a preview of a static raster or vector collection. Whether UNR chooses GeoBlacklight, an API-first platform, or ArcGIS, the requirement for GeoServer should be evaluated separately and resource by resource.

Comparative Fit

Option	Best institutional fit	Aardvark support	Cloud-optimized asset fit	Operational responsibility	Maturity
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ArcGIS Hub	ArcGIS-centered publishing and public data	Custom crosswalk	Mixed; custom integration may be needed	Low to moderate	High
ArcGIS Enterprise Sites	Existing self-hosted ArcGIS Enterprise program	Custom crosswalk	Primarily ArcGIS services	High	High
GeoBlacklight	Academic libraries and geospatial collections	Native	Strong	Moderate to high	High
OGM API with ogm-discovery	API-first, reusable library discovery	Native	Strong	Moderate	Emerging
CKAN	Broad institutional or government open data	Custom schema or crosswalk	Extension-dependent	Moderate to high	High

Recommendation for the University of Nevada, Reno

UNR should pursue an API-first, OpenGeoMetadata-compatible architecture as its preferred future direction. The OpenGeoMetadata API, a configurable frontend such as [ogm-discovery](#), and S3-compatible cloud storage align most closely with the metadata and asset recommendations in this report.

This model would allow UNR to:

- Maintain Aardvark as its canonical discovery schema.
- Deliver IIIF, COG, PMTiles, and other assets directly from object storage.
- Avoid making GeoServer a universal production dependency.
- Reuse the same catalog through a website, QGIS, scripts, and AI-assisted research tools.
- Replace or redesign the public interface without rebuilding the metadata service.
- Participate directly in the OpenGeoMetadata community's emerging shared infrastructure.

This recommendation carries an early-adopter obligation. UNR should expect to contribute testing, documentation, interface requirements, and possibly development effort. Before the architecture is treated as production infrastructure, UNR should confirm a named service operator, supported API versions, change and deprecation procedures, incident communications, metadata-harvest expectations, backup and restoration responsibilities, complete metadata export, and independent retrieval of its assets.

GeoBlacklight is the strongest conservative alternative. If UNR wants a mature academic-library application with a substantial implementation community—and can commit to supporting Ruby on Rails and Solr—GeoBlacklight paired with S3-compatible storage would be an excellent and lower-risk choice. UNR should select that alternative if the shared API cannot demonstrate acceptable governance, service continuity, security ownership, metadata export, independent asset retrieval, and an actionable exit procedure by the production decision gate. A new GeoServer dependency would not be required.

ArcGIS Hub should remain under consideration if UNR's central GIS organization is prepared to own the portal and the university has already standardized public-data publishing around ArcGIS Online. Before selecting it, UNR should test representative Aardvark records, historical maps, IIIF resources, COGs, PMTiles, compound objects, and archival relationships. The institution should also confirm that metadata can be exported without loss.

CKAN is not recommended as the primary platform for this project unless the scope expands into a general university open-data program.

Regardless of the selected interface, UNR should preserve a vendor-neutral foundation. Canonical metadata, persistent identifiers, archival files, and cloud-optimized derivatives should remain independently accessible and exportable. That foundation will make the geoportal replaceable—which is one of the best indicators that the collections themselves are being managed for the long term.

Estimated Costs

Section Summary

UNR should establish a provisional five-year budget of \$300,000 for the recommended repository program: \$110,000 for implementation, \$140,000 for ongoing operations, and \$50,000 in contingency. The principal investment is staff and specialist time to prepare collections, establish reliable publishing workflows, and sustain the service. Storage and other direct operating services account for a comparatively small share of the budget. A representative pilot, funded within the implementation allowance, should establish the workload and staffing commitments needed to refine these estimates.

Basis of the Estimate

The recommended architecture combines a shared OpenGeoMetadata API, a configurable discovery interface, GitHub-based metadata management, and object storage for public access files. This approach allows UNR to direct its investment toward collection stewardship and local service delivery while participating in shared infrastructure. Its financial sustainability depends on clear responsibilities for both the shared services and the work retained within the Libraries.

The estimates below are planning assumptions based on defined quantities of work. They are not vendor quotations or an assessment of current UNR staffing costs. The model assumes an initial collection of 25,000 existing metadata records and 5 TB of public access derivatives, followed by approximately 1,000 additional records and 1 TB of storage growth annually after launch. The collection inventory and pilot should confirm these quantities.

The initial collection is assumed to support predominantly automated metadata conversion and batch preparation of access files, with selective review of exceptions. Included activities cover collection assessment, rights and publishing responsibilities, metadata conversion, asset processing, interface configuration, accessibility testing, documentation, and staff training. Original files remain within UNR's preservation environment. Large-scale digitization, conservation, extensive manual georeferencing, new preservation infrastructure, and restricted-access services would require separate project scopes.

All amounts are expressed in constant 2026 dollars, without inflation. The five-year period begins with implementation and includes a full operating allowance in each year.

Five-Year Program Budget

Budget Component	Basis	Five-Year Amount
Initial implementation	Collection preparation, publishing workflows, and service launch	\$110,000
Continuing operations	\$28,000 annually for five years	\$140,000
Planned expenditure	Implementation + operations	\$250,000
Contingency	20% of planned expenditure	\$50,000
Total planning provision	Five years, including contingency	\$300,000

The first year includes \$110,000 in implementation work and \$28,000 in operating support. Years two through five each carry a \$28,000 operating allowance. The contingency provides capacity to address additional effort within the approved scope, including processing exceptions and remediation identified during implementation.

Initial Implementation

The implementation budget supports four areas of work. Collection assessment, metadata preparation, and asset workflows account for \$86,000. The remaining \$24,000 supports configuration and launch of the public interface. These investments build on the prototype and establish repeatable practices for future collection growth.

Area of Work	Principal Deliverables	Planning Basis	Amount
Collection assessment and governance	Inventory, collection priorities, rights categories, and assigned publishing responsibilities	160 hours at \$100	\$16,000
Metadata preparation	Crosswalks, automated validation, link checking, deduplication,	400 hours at \$75	\$30,000

Area of Work	Principal Deliverables	Planning Basis	Amount
	exception review, and documentation		
Access-file preparation	Reusable IIIF, COG, and PMTiles workflows; batch processing; validation and publication	240 hours at \$150, plus \$4,000 for temporary computing and transfer	\$40,000
Interface configuration and launch	UNR presentation, accessibility and browser testing, staff training, and operational handoff	160 hours at \$150	\$24,000
Total implementation		960 labor hours and \$4,000 in direct services	\$110,000

The hourly rates represent planning allowances for different responsibilities. A final funding request should use UNR's salary and benefit costs for assigned employees and quoted rates for contracted services. Work completed through the planning grant and reusable prototype components should inform the remaining implementation scope.

Metadata condition is a major determinant of cost. The 400-hour allowance assumes that automated conversion and validation can handle most records, leaving staff to review selected records and resolve exceptions. A collection requiring individual descriptive work across most records would need a larger staffing allocation or a phased release schedule. Asset preparation carries a similar dependency: consistent source files support efficient batch processing, while difficult scans, incomplete georeferencing, and collection-specific formats require additional specialist attention.

Pilot and Budget Refinement

The first **\$15,000 of the implementation budget** should support a pilot using up to 500 representative records and samples of each major asset type. The pilot should measure metadata exceptions, hands-on processing time, storage requirements, and accessibility work. It should also establish who will maintain the workflows and respond to routine service issues.

These findings will allow the Libraries to refine the remaining implementation budget before committing to the full collection. The pilot allocation is included in the \$110,000 implementation total.

Continuing Operations and Staffing

An annual allowance of **\$28,000** supports routine collection stewardship, technical maintenance, and operating services. This level of support assumes established workflows, limited local customization, and a shared API whose operator maintains the backend service.

Responsibility	Annual Planning Basis	Annual Amount
Metadata and collection stewardship	240 hours at \$75 for additions, record corrections, link review, publisher coordination, and collection inquiries	\$18,000
Technical maintenance	40 hours at \$150 for dependency updates, workflow repairs, monitoring review, and coordination with the shared service	\$6,000
Storage and operating services	Average allowance for public access files, a recovery copy, requests, monitoring, and related tools	\$4,000
Total annual operations	280 labor hours and operating services	\$28,000

The staffing allowance represents approximately five to six hours per week across the participating roles. The technical portion is limited to 40 hours annually, making access to dependable shared-service support particularly important. Major upgrades, sustained incidents, new integrations, or a formal response-time commitment would require additional technical capacity.

Across five years, the planned \$250,000 expenditure comprises **\$226,000 in labor and \$24,000 in direct services**, before reserving a \$50,000 contingency. Where existing employees undertake this work, part of the program budget represents an allocation of institutional capacity. The funding plan should distinguish that commitment from new expenditures for contracted work, additional staffing, or purchased services. Both require an explicit commitment from the responsible departments.

Storage, Hosting, and Shared Services

Public access files should remain in object storage, with preservation responsibilities governed through UNR's institutional arrangements. As a published pricing benchmark, Cloudflare R2 Standard storage costs \$0.015 per GB-month and does not carry an internet-egress charge.¹ This provides a basis for estimating storage expenditure; provider selection should also consider campus procurement, recovery requirements, and operational support.

The model assumes 5, 6, 7, 8, and 9 TB of public access files in successive years, averaging 7 TB across the period. A public copy and a separately governed recovery copy produce an average billable volume of 14 TB. Using 1 TB = 1,000 GB, the annual service allowance is calculated as follows:

Service	Basis	Average Annual Amount
Storage	14,000 GB × \$0.015 × 12 months	\$2,520
Storage requests	Usage allowance, subject to pilot measurements	\$480
Monitoring and related tools	Planning allowance	\$1,000
Total		\$4,000

The recovery copy protects access derivatives and does not replace preservation of original collection materials. The calculation excludes free-tier discounts. Request and tooling allowances should be refined using measured usage and the services selected for production.

The base budget assumes that the shared OpenGeoMetadata API and frontend hosting are available without mandatory service fees. Before implementation, UNR should confirm the API's operating responsibilities, participation terms, support arrangements, and continuity provisions. Any required contribution or support agreement should become an explicit recurring budget item.

GitHub Pages can support the prototype and may remain suitable for a modest static frontend, subject to its published site-size and bandwidth limits.² Production hosting should be selected after assessing expected traffic, accessibility, indexing, and institutional service requirements. Large collection files and API responses served from other infrastructure do not consume the

¹Cloudflare, "[Pricing.](#)" R2 documentation, accessed September 17, 2026. Standard storage is \$0.015 per GB-month; Class A operations are \$4.50 per million and Class B operations are \$0.36 per million. Internet egress is free; other services may incur separate charges.

²GitHub Docs, "[What is GitHub Pages?](#)" and "[GitHub Pages limits](#)", accessed September 17, 2026. Availability depends on account and repository type. Published sites have a 1 GB size limit and a soft bandwidth limit of 100 GB per month.

frontend host's transfer allowance. Any paid production-hosting requirement should be incorporated into the operating budget.

Collection Growth and Expanded Services

A larger collection program would require additional implementation work and continuing staff capacity. The following illustrative scenario shows how a **\$650,000 five-year program** could be allocated. It adds \$290,000 in planned work and services to the base program, together with \$60,000 in additional contingency. Each addition corresponds to a defined increase in service or collection activity.

Program Expansion	Additional Work or Capacity	Five-Year Increase
Initial metadata remediation	800 additional hours at \$75 for records requiring more extensive review and correction	\$60,000
Initial asset preparation	400 additional hours at \$150 for difficult source files and additional processing workflows	\$60,000
Integrations and service development	300 additional hours at \$150 for separately specified features and connections to other systems	\$45,000
Continuing collection stewardship	An additional \$20,000 annually for intake, review, and publisher support	\$100,000
Infrastructure and service support	An additional \$5,000 annually for storage, traffic, or purchased support	\$25,000
Additional planned expenditure		\$290,000
Additional contingency	20% of the additional planned expenditure, rounded upward	\$60,000
Total increase over the base \$300,000 program		\$350,000

Under this scenario, total planned expenditure would be \$540,000, with \$110,000 in contingency. Of the \$350,000 increase, \$265,000 funds additional labor, \$25,000 funds infrastructure or services, and \$60,000 provides reserve capacity. The expanded program is an illustrative scope option; the collection inventory and service priorities should determine which additions are warranted.

Factors Affecting the Budget

The most consequential variables are the extent of manual metadata work, the condition of source files, and the level of continuing service support. Each additional 100 hours of metadata work adds \$7,500 at the modeled rate; 100 hours of specialist technical work adds \$15,000. A 10% change in the base labor rates changes planned expenditure by \$22,600 before contingency. A support agreement costing \$5,000 annually adds \$25,000 over the five-year period.

Storage growth is more readily calculated. At the benchmark rate, each additional TB retained throughout all five years costs \$900 per copy, or \$1,800 with the modeled recovery copy, before requests and other services. Providers that charge for outbound transfer would also require a traffic allowance.

The Libraries should review these assumptions after the pilot and during annual planning. Contingency supports uncertainty within the approved work; new collections, features, or service commitments should carry their own scope and funding decisions.

Costs of Alternative Platforms

UNR's investment in metadata preparation, access files, and collection stewardship will support any of the platforms considered in this report. Platform selection primarily changes who operates the software, what technical support the Libraries must provide, and whether additional licensing or infrastructure is required. The completed OGM prototype gives UNR a practical starting point and reduces the work remaining to establish a production service.

The following comparison applies the same collection budget to each option: \$86,000 for initial collection preparation and \$22,000 annually for metadata stewardship, access-file storage, and related services. This common five-year investment totals \$196,000 before platform costs and contingency. The estimates assume predominantly automated collection processing, public access, external object storage, and limited local customization.

Platform	Implementation	Annual Support Services	Annual Licensing	Five-Year Program Cost
Shared OGM API and OGM Discovery	6,000–12,000	6,000–14,000	\$0	\$278,000–334,000
ArcGIS Hub Basic	12,000–24,000	6,000–14,000	\$1,400	\$294,000–356,000
Dedicated OGM API and OGM Discovery	6,000–12,000	14,000–22,000	\$0	\$326,000–382,000
GeoBlacklight with object storage	12,000–24,000	14,000–22,000	\$0	\$334,000–396,000
CKAN with object storage	12,000–24,000	14,000–22,000	\$0	\$334,000–396,000
ArcGIS Hub Premium	12,000–24,000	6,000–14,000	\$11,400	\$354,000–416,000
New ArcGIS Enterprise Standard deployment with Enterprise Sites	18,000–36,000	16,000–28,000	\$12,006	\$425,000–518,000

Five-year totals include the common \$196,000 collection investment, remaining platform implementation, five full years of platform support and licensing, and twenty percent contingency. Totals are rounded to the nearest \$1,000. The other columns exclude collection costs and contingency. These are planning allowances, with no inflation adjustment; actual expenditures should reflect confirmed work, institutional support, and current quotations.

Shared OpenGeoMetadata API. The shared-service approach directs most of UNR’s investment toward its collections and publishing workflows. The community operates the backend, while the Libraries maintain their metadata, access files, and public interface. The implementation allowance covers unfinished production checks, accessibility review, recovery verification, documentation, and handoff. This option depends on a confirmed community operating commitment and clear arrangements for support and service continuity.

Dedicated OpenGeoMetadata API. A dedicated deployment can reuse the existing software and prototype work. Its principal additional cost is operating the backend: maintaining the

database and search services, monitoring processing, applying updates, and managing backups and recovery. The comparison allows an additional \$8,000 annually for these responsibilities, producing a five-year difference of \$48,000 including contingency. Any work needed to move the prototype into a different production environment should be scoped separately.

GeoBlacklight with object storage. GeoBlacklight remains a mature alternative for academic geospatial discovery. Its value to UNR would depend on access to reliable Rails and Solr support and the extent to which a standard deployment meets local requirements. Object storage can support routine asset delivery without requiring a separate GeoServer service. The estimate covers a modest implementation and continuing maintenance; substantial interface development or custom integrations would require additional funding.

ArcGIS Hub Basic and Premium. Hub may be competitive where campus GIS staff already provide licensing, templates, and operational support. The table uses historical 2025 public-sector pricing: two Creator accounts at \$700 each annually, plus a \$10,000 annual Premium subscription where purchased separately. Esri includes Hub Premium in Educational Site/Institution licenses, so UNR should confirm its entitlement before budgeting an additional license expense. If the modeled licenses are fully covered, either Hub option falls to approximately 286,000–348,000 over five years.

The Hub estimates retain large access files in external object storage. Extensive ArcGIS-hosted imagery or feature storage would require a separate credit calculation. UNR should also test metadata conversion, previews, downloads, and export workflows against its collection requirements. Premium's engagement features require additional staff capacity only if the Libraries choose to use them.

CKAN with object storage. CKAN warrants consideration where UNR can participate in an established university open-data service. A dedicated library implementation would require configuration and testing of Aardvark mappings, geospatial previews, and asset workflows. The table assigns CKAN the same implementation and support allowances as GeoBlacklight; the actual difference should be established through a defined scope or service proposal.

ArcGIS Enterprise Sites. A new Enterprise deployment requires infrastructure and an assigned operator in addition to licensing. The table uses a historical annual Standard subscription benchmark of \$12,006 for up to four cores. Campus-covered licensing would reduce the estimate to approximately 353,000–446,000, although administration and infrastructure would remain. If UNR already operates a supported Enterprise environment, the Libraries should assess its actual onboarding requirements and service charges before considering a separate deployment.

The shared OGM approach remains the recommended starting point because it builds on completed work, supports UNR's metadata strategy, and distributes backend responsibilities. Campus-supported alternatives may be financially competitive. The final decision should rest on demonstrated collection support and an operating commitment the UNR Libraries can sustain.

Cost Risks

Collection condition and staff capacity will have the greatest influence on the final budget. The estimates assume that most metadata and access files can be prepared through automated workflows, with staff reviewing exceptions. Incomplete descriptions, inconsistent identifiers, unresolved rights, difficult scans, or extensive manual georeferencing could increase the work substantially. A representative collection sample should establish processing effort before UNR commits to a full publication schedule. At the planning rates used here, each additional 100 hours of metadata work adds \$7,500, while 100 hours of specialist technical work adds \$15,000, before contingency.

Continuing support must be assigned explicitly. The shared OGM estimate provides approximately 40–80 technical hours annually; locally operated platforms provide approximately 80–120 hours. These allowances support routine maintenance and limited troubleshooting. Major upgrades, sustained incidents, extensive customization, or round-the-clock response would require additional capacity. Existing staff can provide some of this support, but their time must be committed alongside other responsibilities. UNR should also confirm the shared API's operating terms, incident procedures, and continuity arrangements before relying on it for production service.

Restricted collections and additional services can change the architecture as well as the budget. Authentication, controlled downloads, signed asset links, and audit records should be scoped against specific access requirements and available campus services. GeoServer, synchronized legacy systems, and new integrations should likewise be introduced only where a defined need justifies their continuing operational cost. The public-access baseline does not include these expanded services, large-scale digitization, or new preservation infrastructure.

Storage growth, download traffic, ArcGIS credits, licensing changes, and purchased support can increase annual expenses. UNR should confirm campus coverage and departmental charges, measure representative usage, and review actual costs each year. Every additional \$1,000 in annual expense adds \$6,000 to the five-year budget when the twenty percent contingency is included.

The comparison already includes **twenty percent contingency**. UNR should retain that provision until collection quantities, remaining production work, and support commitments are confirmed. Additional collections or features should carry separate estimates and funding decisions. This approach allows the Libraries to expand the service deliberately while preserving the metadata and asset foundation that supports future platform changes.

Public-Sector Cost Evidence

Public records can document the scale of comparable initiatives while clarifying the limits of a direct comparison. Minnesota IT Services reports **\$975,900 in approved Technology Modernization Fund support** for “Modernization of the Geospatial Commons Platform,”

approved February 8, 2024. The published scope includes a transition from on-premises infrastructure to Azure, security improvements, and expanded web-service delivery.³

This amount represents an approved modernization allocation. The public project summary does not separately identify payments to Esri, licensing, internal labor, cloud services, or final expenditure. It therefore provides context for the scale of a statewide modernization effort, but does not establish a price for migration from CKAN to ArcGIS or for UNR's implementation. Minnesota also distributes enterprise GIS licensing and Commons costs across participating state agencies, further limiting a direct institutional comparison.⁴

An itemized comparison would require project contracts, expenditure records, and a separation of implementation costs from continuing services. For UNR's planning, published service prices support individual calculations, while local workload measurements and staffing commitments remain the basis for the program estimate.

Recommendation for the University of Nevada, Reno

UNR should use **\$300,000, including contingency, as the provisional five-year planning provision** for the defined public-access program. The Libraries should begin with the \$15,000 pilot included in that amount, then refine implementation and operating commitments using the collection inventory, measured processing effort, and confirmed shared-service arrangements.

The funding plan should assign responsibility for metadata stewardship, asset workflows, accessibility, and technical maintenance, identifying both existing staff capacity and any new expenditure. Collection preparation and documented workflows should remain priorities because they support continuity across future platform changes. Additional collections or services should be introduced through explicit scope decisions, with the corresponding implementation and recurring costs identified at the time of approval.

³Minnesota IT Services, "[Modernization of the Geospatial Commons Platform](#)", accessed September 17, 2026. The project record lists approval on February 8, 2024, and approved funds of \$975,900.

⁴Minnesota IT Services, "[Geospatial](#)", accessed September 17, 2026. Describes enterprise GIS offerings and shared costs across state agencies.

Prototype

Section Summary

The working prototype demonstrates that UNR can build a modern geoportal using the same API-and-frontend architecture developed for the Big Ten Academic Alliance. The OpenGeoMetadata API powers search and previews, the customizable React interface delivers the public experience, and OpenGeoMetadata Studio supports the metadata lifecycle. Together, they validate the recommended approach and provide a practical foundation for a production pilot.

The Statement of Work calls for a focused prototype demonstrating platform selection, data ingestion, metadata display, and search functionality. The University of Nevada, Reno geoportal prototype fulfills that requirement through a working implementation of the architecture recommended in this report.

The prototype now demonstrates more than a public discovery interface. It represents an end-to-end metadata and access lifecycle extending from the preparation of an archival or geospatial resource through metadata creation, derivative generation, versioned publication, API indexing, public discovery, object preview, download, citation, and external reuse.

Three applications perform distinct roles within this lifecycle:

- OpenGeoMetadata Studio supports metadata preparation, validation, enrichment, distribution management, asset processing, and publication.
- The OpenGeoMetadata API harvests, normalizes, indexes, interprets, and exposes the resulting records.
- ogm-discovery provides the UNR-branded public search and discovery interface.

The metadata, assets, API, and public frontend remain independently managed. This is the prototype's most important architectural property. No application becomes the only viable home of the collection.

Prototype Architecture

The prototype combines six principal components:

Component	Role
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OpenGeoMetadata Studio	Imports, edits, validates, enriches, and exports Aardvark metadata; manages typed distributions; and coordinates the creation of cloud-optimized derivatives and supplemental provenance.
GitHub	Maintains UNR's portable, version-controlled Aardvark discovery records in the OpenGeoMetadata/edu.unr repository.
Amazon AWS S3	Delivers original files, COGs, PMTiles, IIIF resources, GeoJSON, GeoParquet, thumbnails, static maps, manifests, and supplemental documentation.
OpenGeoMetadata API	Harvests, indexes, searches, interprets, and exposes the metadata through documented endpoints.
OpenGeoMetadata Discovery	Provides the branded React frontend, search interface, maps, resource pages, and asset viewers.
GitHub Pages	Hosts the compiled frontend and Metadata Studio application as static files without requiring a UNR-operated public application server.

This division reflects the report's central architectural recommendation: metadata creation, preservation, asset processing, versioned publication, discovery services, and the public interface should remain related but independently manageable.

Prototype Metadata and Asset Lifecycle

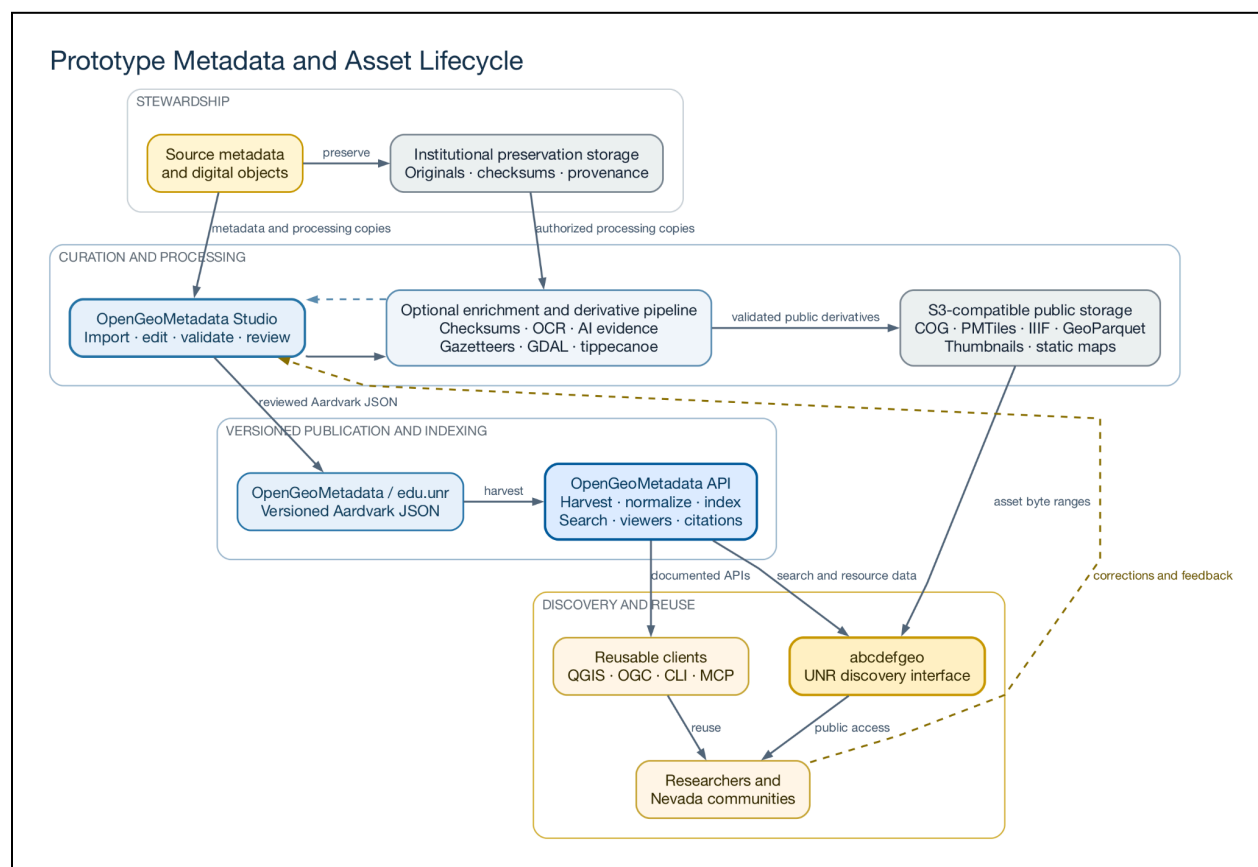
The prototype's end-to-end data flow is as follows:

1. Existing Aardvark records, CSV files, JSON files, DuckDB backups, or records harvested from an OpenGeoMetadata GitHub repository enter OpenGeoMetadata Studio.
2. Studio loads the records into a browser-native DuckDB workspace, where staff can search, inspect, edit, and validate the metadata.
3. Curators review required, descriptive, provenance, object, administrative, relationship, rights, and spatial fields through an Aardvark-aware editing interface.
4. Typed distributions connect each record to its archival original, access derivatives, presentation assets, manifests, supplemental metadata, and enrichment provenance.
5. When appropriate, the optional enrichment pipeline analyzes source imagery or geospatial packages, computes checksums, creates cloud-optimized derivatives, performs OCR and metadata extraction, and prepares an Aardvark draft for human review.
6. Reviewed metadata is exported as individual Aardvark JSON records. Parquet files and a DuckDB snapshot provide additional publication and backup representations.

7. Published JSON records are committed to the OpenGeoMetadata UNR repository, preserving a versioned representation independent of the editing application.
8. The OpenGeoMetadata API harvests, normalizes, and indexes the published records.
9. The ogm-discovery frontend requests search results, record displays, viewer configurations, citations, relationships, and download information from the API.
10. COG, PMTiles, IIIF, thumbnail, static-map, and other asset requests are delivered directly from object storage.
11. Corrections and enrichment decisions return to the curation layer, allowing records to be reviewed, republished, and reindexed without reconstructing the collection inside the public geoportal.

The archival originals remain outside this discovery lifecycle in institutionally managed preservation storage. Metadata Studio and the public asset store operate upon controlled access or processing copies. The public derivatives can be regenerated from preserved originals through documented workflows.

Figure 1. Prototype metadata and asset lifecycle. Source metadata and digital objects move through preservation, Metadata Studio curation, derivative generation, versioned OpenGeoMetadata publication, API indexing, and the UNR-branded public interface. Corrections return to the curation layer rather than being stored only in the public application.



UNR-Branded Public Geoportal

The public geoportal prototype is available at:

<https://opengeometadata.org/ogm-discovery/unr/#/>

The University of Nevada, Reno theme should be enabled by default via the link above, or it can be selected from the theme menu in the lower-right corner of the application. The underlying source code is maintained in the [ogm-discovery repository](#).

The prototype demonstrates that a shared discovery application can be presented as a recognizable UNR service without forking or duplicating the underlying search infrastructure.

The UNR theme supplies:

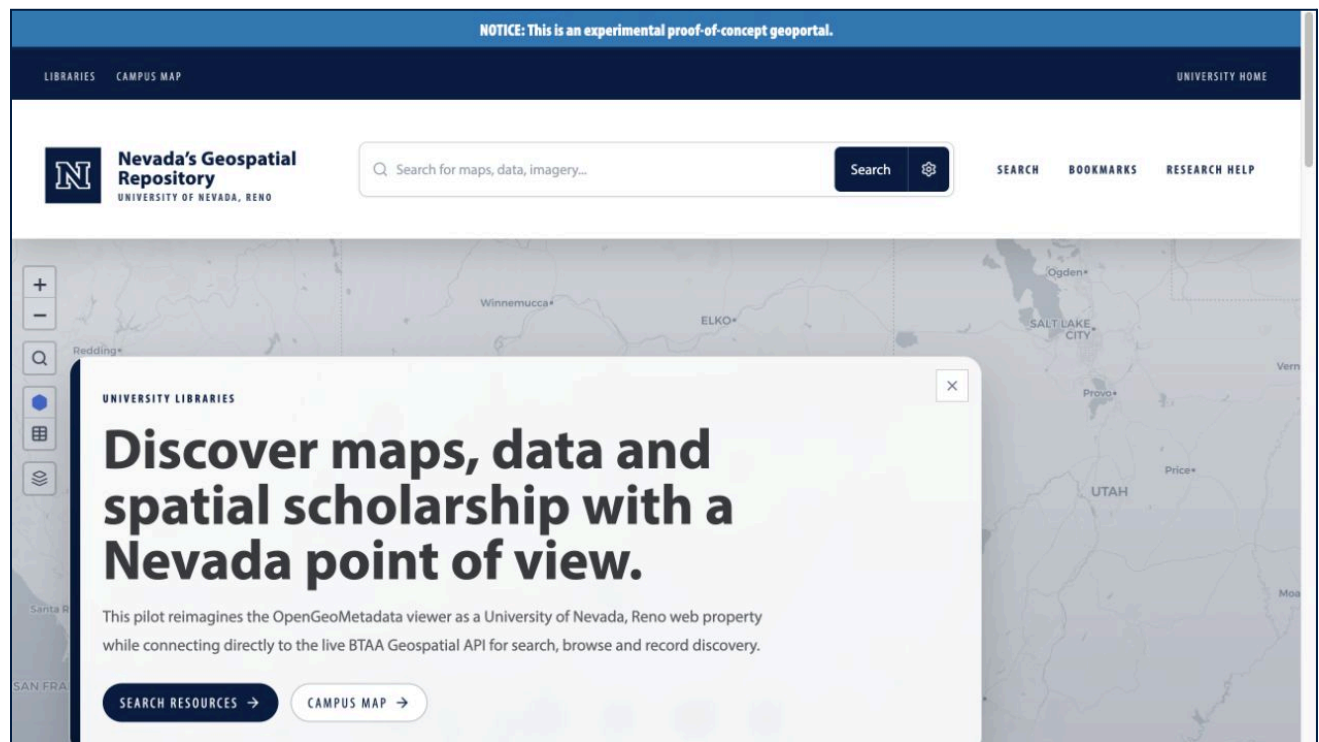
- University logo and wordmark treatment
- UNR colors and typography
- University Libraries and campus navigation
- Institutional footer and contact links
- A proof-of-concept notice
- Nevada-centered map positioning
- UNR-specific featured resources
- Curated links for Nevada, Great Basin, watershed, land-use, and historical-map discovery
- UNR-specific favicon and progressive-web-application assets
- An API filter restricting discovery to UNR records

These settings are defined in the frontend's `themes/unr.yaml` configuration rather than scattered across application components. The same React application can consequently support multiple institutions while preserving a common interface, accessibility model, and API integration.

The UNR theme applies an automatic API filter for the University of Nevada, Reno. Institutional scope is therefore expressed through the documented API contract rather than through client-side filtering after a larger result set has been downloaded.

Figure 2. The UNR-branded prototype homepage, including institutional navigation, a Nevada-centered discovery map, a prominent search interface, featured resources, and

an explicit proof-of-concept notice.



Search and Discovery

The prototype provides working keyword, faceted, temporal, and spatial discovery.

At the time of review, the UNR OpenGeoMetadata repository contained thirteen representative Aardvark records: eight datasets and five maps. These records include scanned historical maps, georeferenced rasters, vector index layers, and related cloud-optimized derivatives.

A search for “Nevada” returns the thirteen representative UNR resources. Results can be presented in map, list, or gallery form and sorted by relevance, year, or title.

The interface provides facets for:

- Place
- Resource class
- Resource type
- Subject
- Theme
- Creator
- Publisher
- Provider

- Access
- Georeferenced status

A year histogram and start/end fields allow temporal filtering. Spatial results are plotted on an interactive map, while separate controls support place searching and geographic exploration. Active filters remain visible and removable.

Include and exclude operations are supported by the API rather than simulated entirely within the browser. Search URLs preserve query and filter state, making searches shareable and allowing users to move backward and forward without losing context.

The prototype also demonstrates:

- Advanced search
- Search suggestions
- Map, list, and gallery views
- Result sorting
- Result bookmarks
- Resource-class shortcuts
- Geographic clustering
- H3 resource-density summaries
- Previous and next navigation among results
- Links from metadata values back into filtered searches
- JSON and CSV result representations

Figure 3. UNR search results for “Nevada,” showing active filters, map/list/gallery controls, a geographic result map, and representative UNR resources.

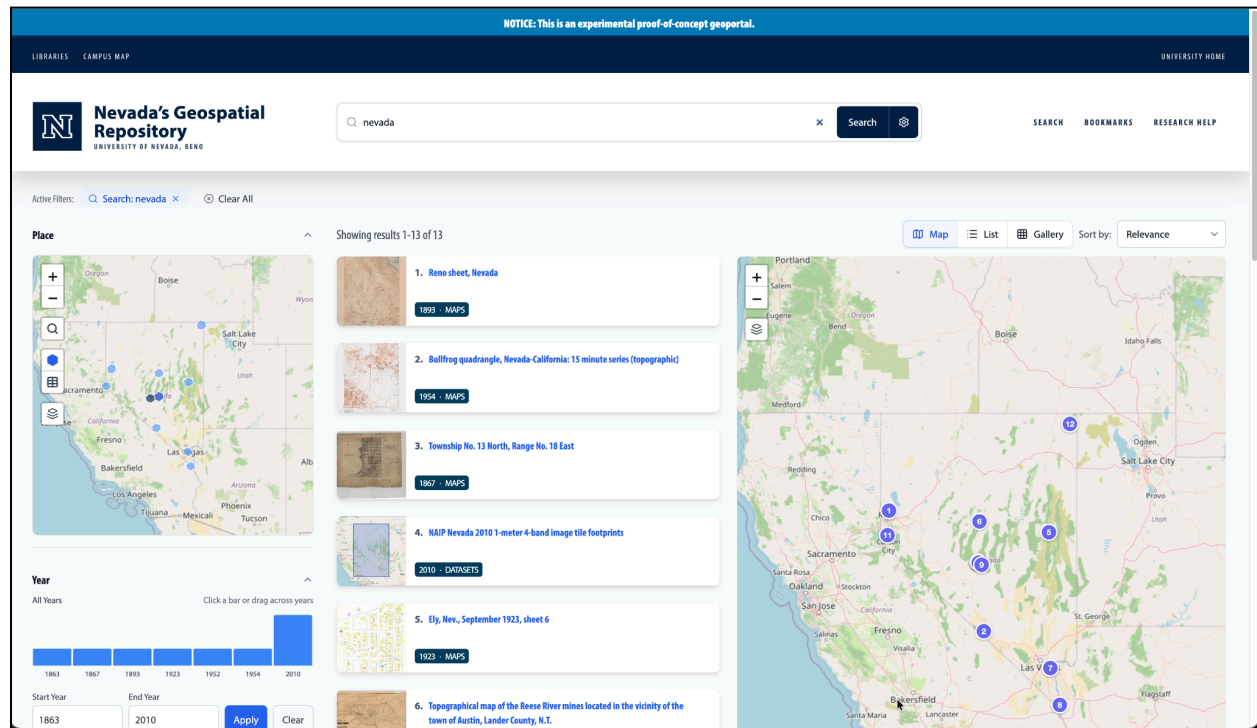


Figure 4. UNR advanced search builder.

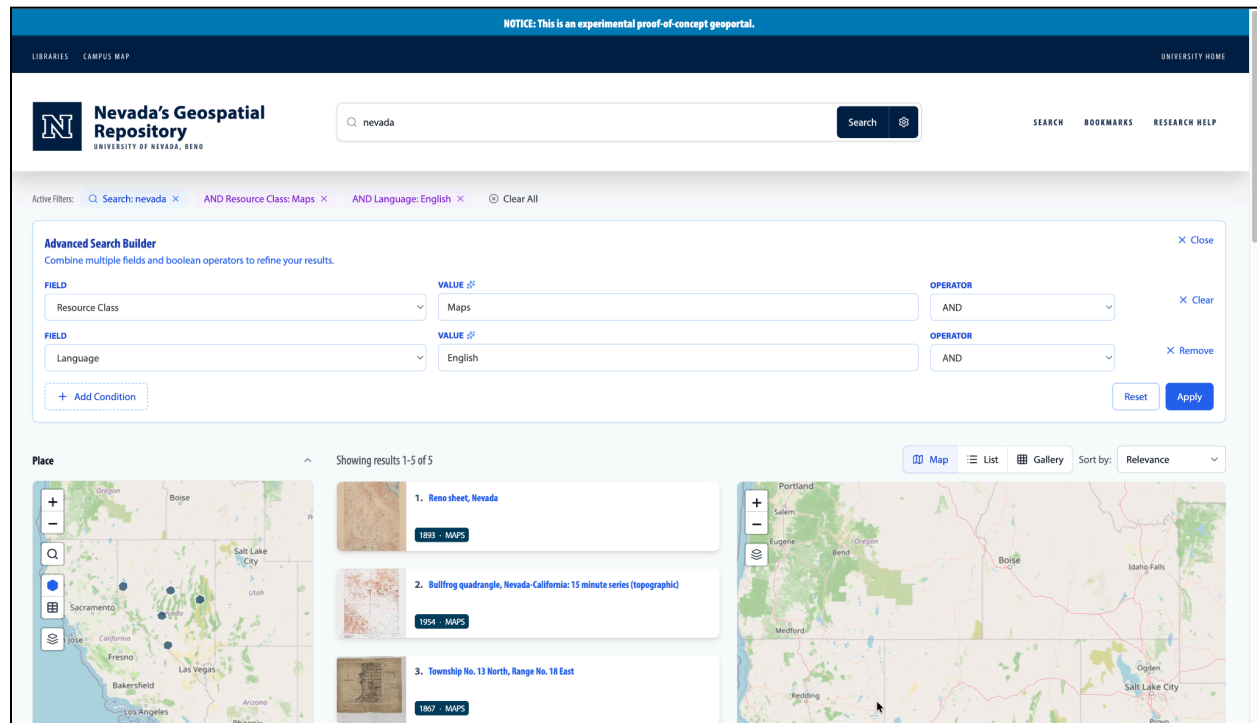
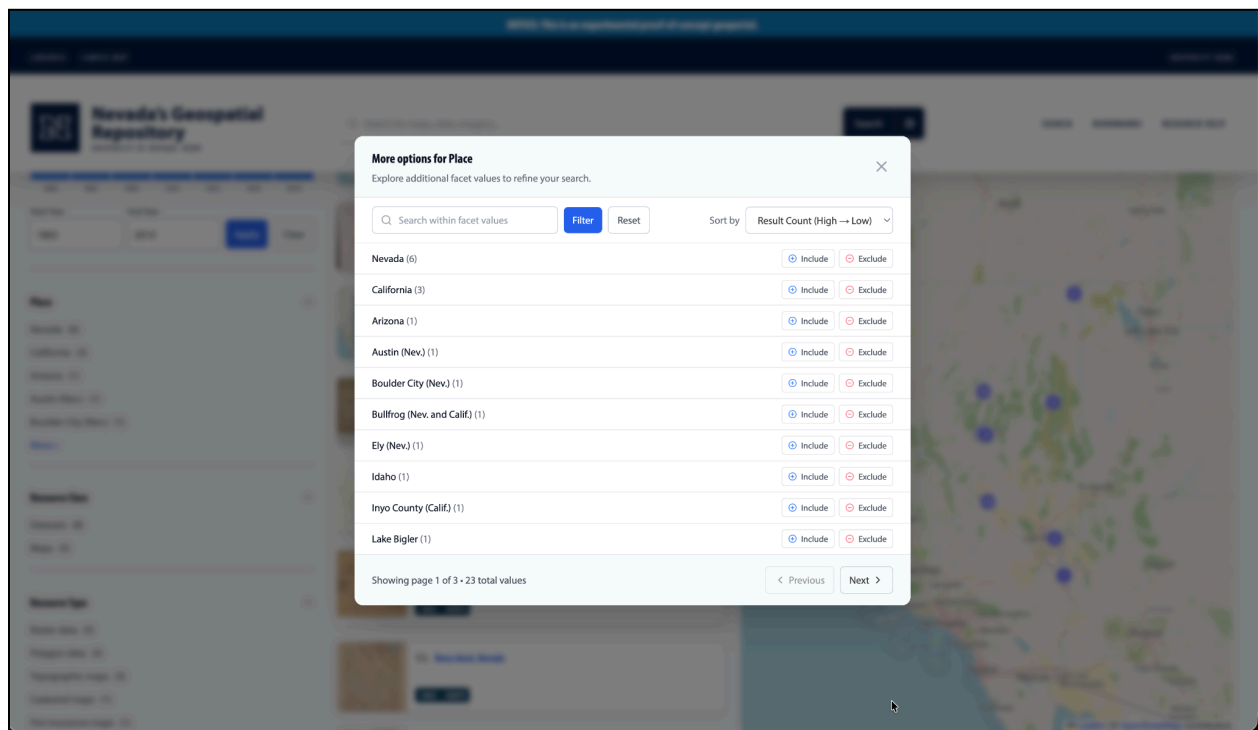


Figure 5. UNR faceting options.



Resource Metadata Display

Resource pages translate Aardvark discovery metadata into a readable public presentation.

The pages include:

- Title and date
- Resource class and resource type
- Description
- Language
- Spatial coverage
- Topic category
- Provider
- Location map
- Processing or use notes
- Original and derivative downloads
- Web-service information
- Citation generation
- RIS, BibTeX, and JSON-LD exports
- Permalink
- Similar items
- Previous and next navigation within a result set

Metadata values remain actionable. A user viewing “Boulder City, 1952,” for example, can follow its resource-type, provider, keyword, or place links back into a filtered search.

The frontend does not need to understand every internal Aardvark field or reconstruct relationships from filenames. The API supplies record-display, viewer, download, citation, relationship, and similar-item representations appropriate to the interface.

Cloud Optimized GeoTIFF Preview

The “Boulder City, 1952” resource demonstrates browser display of a Cloud Optimized GeoTIFF.

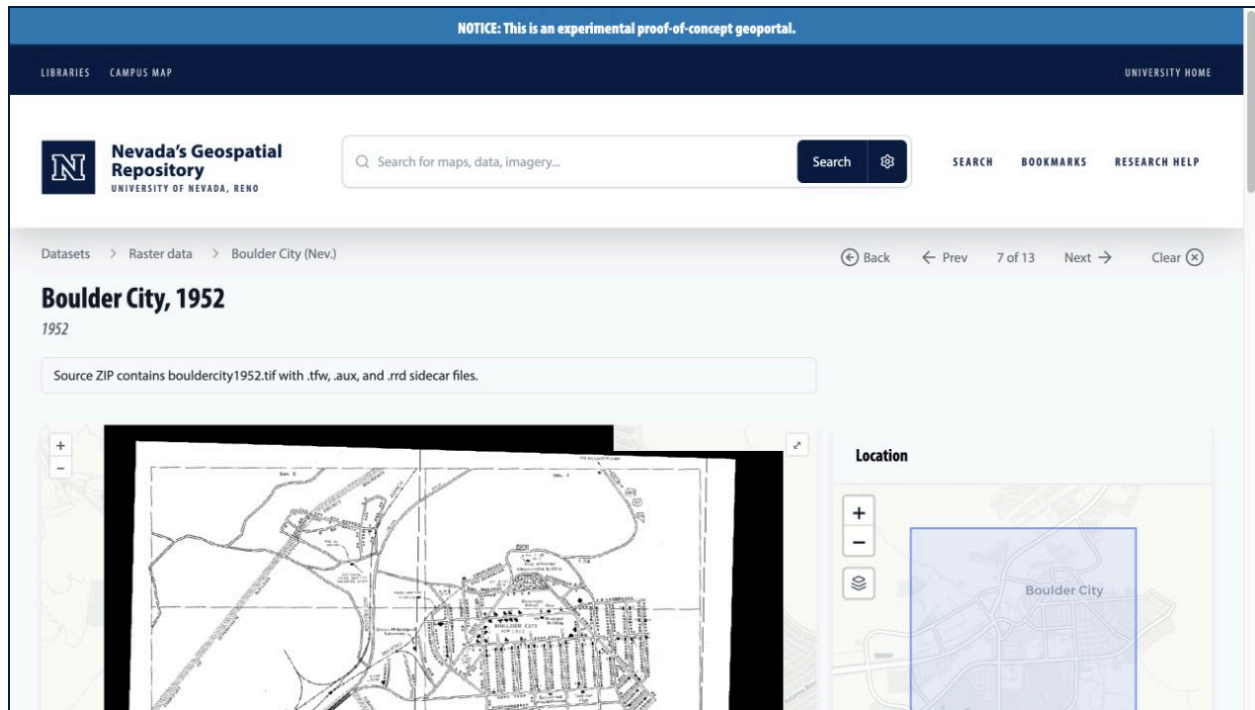
Its Aardvark record identifies:

- A zipped original geospatial raster package
- A COG derivative
- A static map
- A thumbnail
- An Aardvark JSON representation
- An archival accession supplement
- A dataset manifest
- The resource’s geographic bounding box and geometry

The API’s viewer endpoint interprets the typed COG reference and returns a compact viewer configuration containing the COG protocol, object-storage endpoint, and resource geometry. The React frontend uses that representation to display the georeferenced raster beside a location overview.

The browser retrieves only the required byte ranges from the COG rather than downloading and decoding the entire raster before rendering it. This demonstrates the object-storage-first model described in the Cloud-Optimized Assets section.

Figure 6. Resource-level COG preview for “Boulder City, 1952.” The primary viewer displays the raster while the location map communicates its geographic extent.



PMTiles Vector Preview

The “NDOT Georeferenced TIFF Index Polygons” record demonstrates delivery of a vector layer through PMTiles.

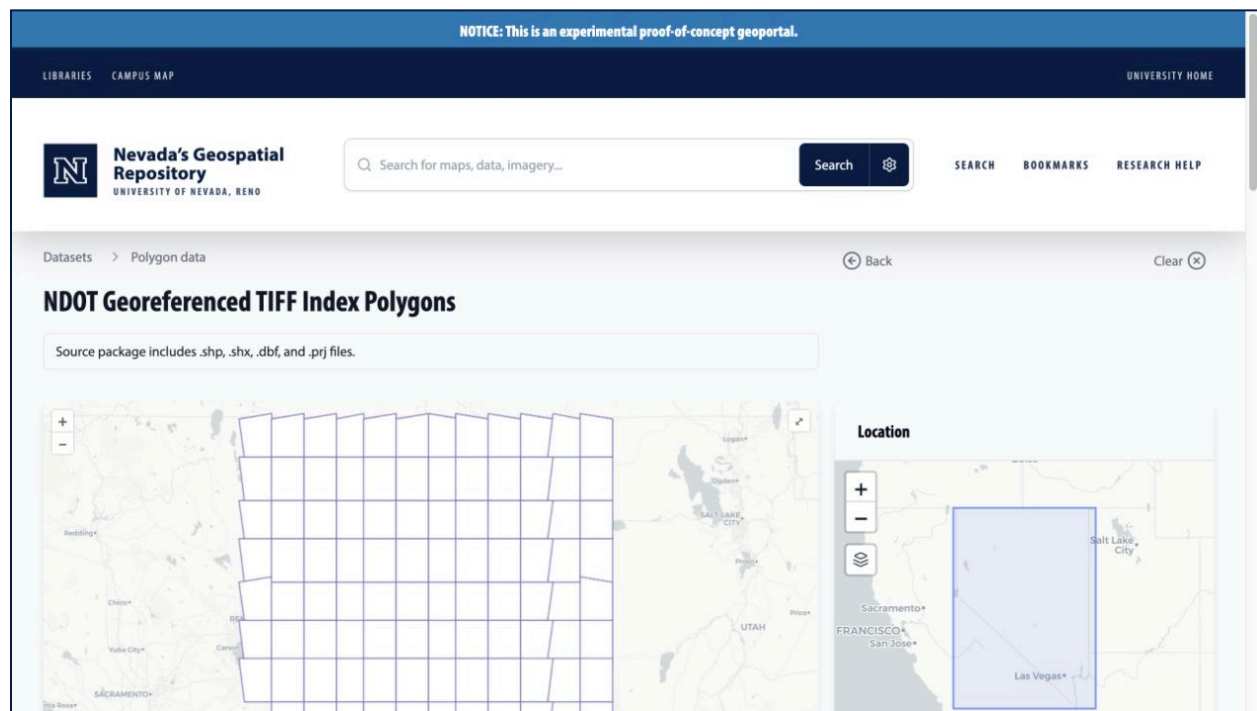
The source Aardvark record identifies:

- The original zipped shapefile
- A GeoParquet derivative
- A PMTiles vector-tile derivative
- A GeoJSON derivative
- A static map and thumbnail
- Supplemental accession documentation
- A statewide bounding geometry

The API recognizes the typed PMTiles reference and returns a viewer configuration identifying the PMTiles protocol, endpoint, and geometry. The frontend then loads the vector tiles directly from object storage.

The preview displays the index polygons without requiring GeoServer, WMS, or a dynamically generated tile service. Original files and alternate analytical formats remain available as downloads.

Figure 7. PMTiles vector preview for the NDOT georeferenced TIFF index. The object layer is retrieved directly from cloud storage and displayed alongside its statewide location extent.



Additional Object Previews

The same pattern supports the other asset types described in this report:

- IIIF for scanned maps and non-georeferenced imagery
- COG for georeferenced raster resources
- PMTiles for tiled vector data
- GeoJSON for smaller vector previews
- Static maps and thumbnails for search results, sharing, accessibility fallbacks, and error states

The OpenGeoMetadata API acts as the interpreter between discovery metadata and the frontend. It determines the available viewer protocol and returns a normalized configuration. The React application can then select the appropriate viewer without embedding institution-specific file paths or guessing asset types from extensions.

This arrangement also contains failures. If a COG, PMTiles, or IIIF preview is unavailable, the resource page can continue presenting its title, metadata, rights, citation, location, and download links.

OpenGeoMetadata Studio

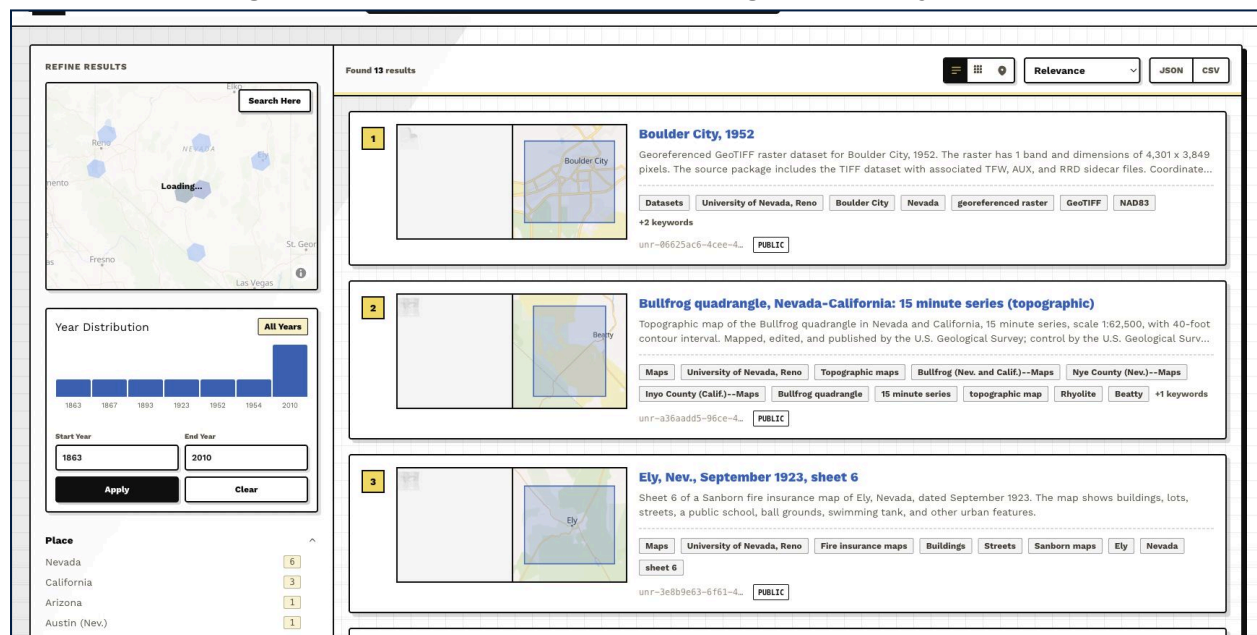
The public prototype demonstrates what happens after UNR metadata has been published: records are harvested, indexed, searched, displayed, and connected to cloud-optimized assets. A sustainable repository also requires an upstream workspace in which staff can prepare those records and manage their relationships to the underlying digital objects.

[OpenGeoMetadata Studio](#) provides that missing curation layer. The project is a browser-native metadata-management workspace developed specifically for the OpenGeoMetadata Aardvark standard. Its [source code and documentation](#) are openly available, and its development and enrichment workflows were presented at [Geo4LibCamp 2026](#).

At the time of review, the authenticated Studio prototype contained the same thirteen representative UNR resources used by the public geoportal prototype. Those records were connected to 135 distribution entries representing original packages, Aardvark JSON, cloud-optimized derivatives, thumbnails, static maps, IIIF resources, archival accession supplements, dataset manifests, and AI-enrichment documents.

This is an important distinction. The metadata record does not merely identify a title and geographic extent. It becomes the organizing point for an entire managed resource package.

Figure 8. The authenticated OpenGeoMetadata Studio workspace contains thirteen representative UNR Aardvark resources. Faceted search, temporal filtering, geographic review, and export controls allow the working metadata collection to be examined before publication through the OpenGeoMetadata API and ogm-discovery.



Browser-Native Metadata Management

OpenGeoMetadata Studio is built with React, TypeScript, Vite, and DuckDB-WASM. Core metadata operations take place in the browser, with records loaded into DuckDB and persisted locally through browser storage. This enables fast SQL-backed searching, filtering, aggregation, and editing without requiring a permanently running metadata-application server.

The authenticated workspace provides:

- Faceted, temporal, and spatial exploration of the working collection
- A searchable administrative resource table
- Aardvark-aware resource creation and editing
- Required, descriptive, provenance, object, administrative, and relationship field groups
- Geographic bounding boxes, geometries, centroids, and georeferenced status
- Rights, licensing, suppression, membership, and version relationships
- A separate distribution-management interface
- JSON and CSV inspection and export
- Google authentication for protected editing workflows

The discovery interface inside Studio is not intended to replace the public geoportal. It gives metadata staff a way to examine the collection as users and downstream systems will encounter it. Facets, maps, record pages, downloads, and citations become quality-control tools as well as discovery features.

For “Boulder City, 1952,” the editor separates fields into functional groups:

- Required metadata
- Identification and description
- Provenance
- Object geometry and technical characteristics
- Administrative metadata and rights
- Relationships
- Distributions and assets

This organization translates the largely flat Aardvark record into an interface that can be understood and maintained by staff without concealing the underlying standard.

Import, Export, and Git-Based Publication

Studio can import Aardvark CSV and JSON files, restore a DuckDB backup, or scan a GitHub repository for `metadata-aardvark` directories. Existing identifiers allow imported records to be updated rather than duplicated.

The workspace can export:

- A complete DuckDB backup
- A ZIP containing individual Aardvark JSON records
- A Parquet representation of the resources
- A companion Parquet representation of distributions

These artifacts can be written into a local repository and committed through an ordinary Git workflow. GitHub then provides version history, review, branching, backup, and a portable exchange mechanism for the published discovery records.

The publication workflow writes resource and distribution data into static artifacts that can be built and delivered through GitHub Pages. The corresponding Aardvark JSON can be committed to an OpenGeoMetadata repository for harvesting by GeoBlacklight, the OpenGeoMetadata API, or another compatible service.

Direct synchronization of every browser edit back to GitHub remains an evolving part of the project. The present prototype therefore treats explicit export, review, commit, and push as a useful publication control rather than claiming that every local change is automatically released.

Distribution and Asset Management

Studio treats distributions as managed relationships rather than as unstructured URLs placed arbitrarily in a record.

For “Boulder City, 1952,” the workspace connects the Aardvark record to:

- The original zipped geospatial raster package
- A Cloud Optimized GeoTIFF
- A thumbnail
- A static map
- The exported Aardvark JSON
- A dataset manifest
- A human-readable archival accession supplement
- A machine-readable archival accession supplement

Each distribution has a relation type, optional label, and URL. Recognized relation identifiers allow downstream systems to distinguish an archival source from a thumbnail, COG, PMTiles archive, IIIF service, static map, or supplemental metadata document.

This typed approach is what allows the OpenGeoMetadata API to choose an appropriate viewer and allows ogm-discovery to present downloads without guessing their meaning from filenames.

The distribution manager also makes missing or inconsistent relationships visible. Staff can search across resource identifiers, titles, relation types, labels, and URLs. This provides a practical quality-assurance surface for identifying broken links, duplicate assets, incomplete derivative sets, and inappropriate relation types before records are published.

Enrichment and Derivative Generation

Studio includes an optional enrichment workbench for transforming source imagery and geospatial packages into reviewed metadata drafts and web-ready derivatives.

The workflow can:

- Accept individual files, folders, imagery, ZIP archives, shapefile components, and georeferenced raster sidecars
- Match companion text, FGDC, ISO, or generic XML metadata
- Compute checksums before processing
- Upload original or processing packages to S3-compatible storage
- Inspect raster and vector packages
- Generate IIIF Level 0 images, thumbnails, and static maps
- Generate COG, GeoJSON, GeoParquet, FlatGeobuf, and PMTiles derivatives when the required tools are available
- Perform OCR and image-based metadata extraction
- Derive reviewable placenames from visible map text
- Compare extracted placenames against local gazetteer indexes
- Produce an Aardvark metadata draft
- Publish the reviewed resource and its distributions into the working catalog

Private storage, OCR, and model operations are mediated through a local proxy. Secrets remain outside the public browser application, and configuration profiles refer to environment-variable names rather than embedding credentials in the interface.

The proxy can use GDAL and tippecanoe for deterministic geospatial inspection and derivative generation. A zipped shapefile, for example, can produce a dataset manifest, GeoJSON preview, GeoParquet access file, and PMTiles presentation layer. A georeferenced raster can produce a COG, while an ordinary scanned map can produce IIIF Level 0 tiles.

Derivative generation is designed to degrade predictably. Missing optional software may prevent a particular derivative from being produced, but the condition is recorded in the dataset manifest instead of preventing the metadata record from being created.

AI-Enrichment Provenance

AI-assisted metadata should not silently replace cataloger judgment. Studio addresses this concern by separating reviewed Aardvark metadata from a companion `ai-enrichments.json` document.

The companion record can preserve:

- Provider and model information

- Prompts and processing parameters
- Redacted requests
- Raw and parsed responses
- Extracted map text
- OCR coordinates and confidence information
- Derived placenames
- Field-level evidence
- Gazetteer candidates and selected matches
- Derivative-generation information
- Indexing suggestions

This separation keeps the Aardvark record focused upon reviewed discovery metadata while retaining evidence about how suggested values were produced. Staff can revise the Aardvark description without discarding the enrichment history, and the enrichment process can be rerun as prompts, models, gazetteers, or normalization rules improve.

Gazetteer reconciliation is intentionally evidence-based. Placenames should be supported by visible map text, OCR coordinates, companion metadata, or another reviewable source. Local indexes can compare candidate labels against Who's On First, OpenStreetMap, GeoNames, GNIS, Wikidata, and a developing canonical OpenGeoMetadata gazetteer without depending upon public geocoding services during every processing run.

AI-derived descriptions, subjects, dates, placenames, and other values remain suggestions until reviewed. The prototype demonstrates how AI can assist metadata preparation without erasing provenance or making an unreviewed model response the permanent catalog record.

OpenGeoMetadata API Backend

The public service at ogm.geo4lib.app is a deployment of the BTAA Geospatial API. Its source code is maintained in the OpenGeoMetadata community's [ogm-api](#) repository .

The API backend uses a contemporary Python service architecture built around FastAPI, Pydantic, PostgreSQL-compatible search infrastructure, Elasticsearch, Redis, and background processing.

Its documented operations include:

- Search
- Advanced search
- Facets
- Suggestions
- Resource records
- Metadata representations
- Viewer configurations
- Downloads and distributions

- Static maps
- Thumbnails
- Citations
- Relationships
- Similar items
- Spatial facets
- H3 map aggregation
- OpenGeoMetadata harvesting
- OGC API–Records
- MCP access

FastAPI supplies a typed application framework and automatically generated OpenAPI documentation. Pydantic models provide a defined contract for requests and responses. Search behavior and metadata interpretation remain on the server, where they can be tested, optimized, secured, and improved for every client.

The API separates canonical records from discovery indexes. Source metadata can be harvested from OpenGeoMetadata repositories, normalized into a shared model, stored in the data layer, and indexed for fast search. Search indexes, thumbnails, static maps, and viewer responses remain reproducible service derivatives rather than the only copies of the metadata.

This is a modern approach because the backend exposes capabilities rather than HTML pages. The same resource, search, citation, or download operation can support a browser, QGIS, a command-line client, another institution’s website, an OGC client, or an MCP-compatible research tool.

React and Vite Frontend

OpenGeoMetadata Discovery ([ogm-discovery](#)) is a React and TypeScript application built with Vite. It is an evolution of the BTAA Geoportal frontend into a configurable, static-site-friendly OpenGeoMetadata viewer.

React provides a component model for search forms, facets, result cards, maps, metadata tables, citations, viewers, and error states. TypeScript gives those components explicit data contracts and helps identify incompatible API changes during development. Vite provides a fast development environment and produces optimized static assets for deployment.

The frontend repository includes:

- Automated unit and component testing
- Code-coverage checks
- ESLint
- Prettier
- Accessibility testing with Axe and Pa11y
- Theme scaffolding

- Site-file generation
- PWA icon and manifest generation
- GitHub Pages deployment automation

Institutional customization is primarily expressed through YAML and static assets. This reduces the need to alter shared search, map, metadata, and accessibility components simply to create a UNR-branded service.

The prototype is deployed through GitHub Pages. Once compiled, its HTML, JavaScript, CSS, icons, and images can be served as static files through ordinary web infrastructure. UNR does not need to operate a dedicated React application server for this proof of concept.

API-First and Backend-for-Frontend Design

The public prototype follows an API-first, Backend-for-Frontend-style arrangement.

Strictly speaking, the GitHub Pages deployment does not operate a separate UNR middleware server. The static React application calls the public OpenGeoMetadata API directly. The API nevertheless performs the BFF role by returning presentation-oriented representations for search, facets, maps, viewers, citations, downloads, and resource pages.

This distinction is useful. UNR receives the advantages of a BFF without introducing another server into the prototype:

- Frontend and backend independence
- Lower frontend complexity
- Shared backend investment
- Replaceable presentation
- Replaceable backend deployment
- Static frontend hosting
- Reduced public attack surface
- Centralized caching and performance work
- Consistent browser, GIS, OGC, command-line, and MCP access
- Institutional reuse through configuration
- Clear testing boundaries
- Metadata and asset portability

Search syntax, metadata normalization, citations, viewer selection, and relationship logic remain behind documented endpoints. Improvements to those functions benefit UNR, the BTAA frontend, and other API clients rather than being implemented separately in every browser application.

A production deployment may choose to place a thin same-origin proxy between the browser and API. That layer could protect server-side credentials, coordinate authentication, apply

institutional rate limits, or avoid cross-origin restrictions. It would not require changing the underlying metadata and asset model.

Any key embedded in a static frontend must be treated as public and rate-limited. Administrative credentials, private asset tokens, and unrestricted API keys must never be compiled into the browser application.

Why OpenGeoMetadata Matters

The prototype works because Metadata Studio, the API, and the public frontend share an understanding of Aardvark metadata and typed resource references.

OpenGeoMetadata provides:

- Stable record identifiers
- Consistent discovery fields
- Spatial bounding boxes and geometry
- Resource classifications
- Provider and institutional scope
- Typed download references
- Typed preview references
- Static-map and thumbnail references
- Rights and access fields
- Relationships among resources
- A community-managed exchange format

Without that shared model, each application would require custom code to interpret local metadata, locate files, decide which viewer to use, and construct citations. Aardvark allows the Studio to author a predictable record, the API to normalize and interpret it, and the frontend to consume an appropriate representation.

Typed references are especially important for object previews. A COG or PMTiles URL is not merely placed in an arbitrary local field. It is identified using a recognized relation URI. The API can therefore return [cog](#), [pmtiles](#), [iiif](#), or another supported protocol intentionally.

This is the difference between a preview that happens to work in one application and an interoperable asset that can be understood across repositories.

Relationship to the BTAA Geoportal

The prototype uses the same general codebase and architectural direction as the Big Ten Academic Alliance's new discovery environment:

- OpenGeoMetadata source records

- Shared harvesting and normalization
- A documented FastAPI backend
- Search and spatial indexes
- React and TypeScript presentation
- API-produced viewer and citation data
- Cloud-optimized assets
- Independent browser, GIS, command-line, OGC, and MCP clients

ogm-discovery generalizes the BTAA frontend into a reusable application that can support additional institutions through configuration. UNR is therefore not beginning with an isolated local codebase. It is testing whether shared community infrastructure can support a distinct institutional geoportal.

OpenGeoMetadata Studio extends that shared direction upstream. It demonstrates how an institution can prepare Aardvark records, manage distributions, generate derivatives, and preserve enrichment evidence before publishing records into shared discovery infrastructure.

This arrangement creates a practical exchange of value. UNR can benefit from improvements developed for the consortium, while its testing of historical maps, Nevada collections, Aardvark editing, COGs, PMTiles, gazetteer reconciliation, and institutional theming can improve the shared software for other adopters.

Prototype Scope and Limitations

The prototype is intentionally labeled as experimental. It demonstrates the recommended direction but should not yet be treated as a production repository.

Its current scope includes:

- Thirteen representative UNR Aardvark records
- 135 typed distribution relationships in Metadata Studio
- UNR-specific branding and homepage content
- Aardvark-aware metadata editing
- GitHub, CSV, JSON, and DuckDB ingestion
- Portable JSON, Parquet, and DuckDB exports
- API-backed search and facets
- Map, list, and gallery results
- Resource metadata display
- COG, PMTiles, and IIIF-capable viewing
- Original and derivative downloads
- Citations and citation exports
- Similar-item discovery
- Static deployment through GitHub Pages
- Optional OCR and AI-assisted metadata drafting
- Dataset manifests and enrichment provenance

- Gazetteer concordance
- Cloud-based derivative generation

It does not yet establish:

- A complete UNR collection inventory
- A production service-level agreement
- Long-term governance of the shared API
- A production custom domain
- Production analytics and monitoring
- A complete metadata remediation workflow
- Restricted-resource authorization
- A production preservation repository
- Final staffing or operational ownership
- A production accessibility conformance report
- A fully automated GitHub publication workflow
- A supported production environment for the enrichment proxy
- Institutional policies governing AI-assisted metadata

GitHub Pages is appropriate for the proof of concept because it demonstrates that the browser applications can be delivered without a complex public server environment. Production hosting may instead use a UNR-managed domain, content-delivery network, managed static host, or server-rendered deployment depending upon requirements for indexing, authentication, analytics, accessibility, and application controls.

Metadata Studio is a curation and processing workspace. The OpenGeoMetadata API is a discovery service. ogm-discovery is a public access layer. None is the preservation repository or the sole authoritative metadata environment.

Recommendation for the University of Nevada, Reno

UNR should accept the expanded prototype as evidence that the recommended metadata-first, API-first, and object-storage-first architecture is technically viable.

The prototype confirms that UNR can:

- Prepare and review Aardvark records through a dedicated metadata workspace.
- Import existing OpenGeoMetadata records without committing them to a proprietary database.
- Connect archival originals, access derivatives, presentation assets, and supplemental metadata through typed relationships.
- Preserve AI-enrichment evidence separately from reviewed discovery metadata.
- Export portable metadata into a version-controlled OpenGeoMetadata repository.
- Publish records through shared OpenGeoMetadata infrastructure.
- Restrict a shared API catalog to UNR resources.

- Present the collection through a distinct institutional theme.
- Search indexed records without operating a local Solr or Elasticsearch service.
- Preview COG and PMTiles objects without operating GeoServer.
- Link preservation-oriented source packages to access and presentation derivatives.
- Generate citations and machine-readable exports through the API.
- Replace or redesign any one application without abandoning the source metadata or assets.

The next phase should use this working lifecycle to answer the remaining production questions: governance of the shared service, metadata workflow ownership, preservation-system integration, accessibility conformance, security review, monitoring, custom-domain hosting, derivative validation, staffing, and recurring funding.

OpenGeoMetadata Studio should be treated as a reference implementation of the curation workflow, not as an irreplaceable system. Its value lies in demonstrating that UNR can maintain a controlled, standards-based path from accessioned files to reviewed metadata, reproducible derivatives, versioned publication, API indexing, and public discovery.

The most important result is not the appearance of any one demonstration interface. It is the confirmation that UNR's metadata, assets, processing tools, API, and public frontend can operate as separate but interoperable layers. That is the architectural property most likely to keep the future repository maintainable as its collections, staff, partnerships, and technologies evolve.

Final Recommendations

Section Summary

UNR should launch a pilot project built on OpenGeoMetadata Aardvark, cloud-optimized assets, the community-sponsored OpenGeoMetadata API, and a lightweight customizable frontend. This approach provides the clearest path to an affordable statewide geoportal—approximately \$300,000 over five years—while preserving UNR’s data portability and avoiding vendor lock-in. Production launch should follow once UNR establishes service ownership, staffing, governance, accessibility, preservation, and partner participation.

The Nevada Historical and Environmental Geospatial Repository should be understood as a statewide stewardship program with a public geoportal—not simply as a website with a collection behind it.

That distinction is central to the recommendations in this report. If UNR begins by selecting a product and then reshapes its collections around that product, it risks reproducing the dependencies that contributed to the retirement of the Nevada Geospatial Data Gateway. If it begins instead with portable metadata, persistent identifiers, validated digital assets, documented workflows, sustainable staffing, and clear institutional ownership, the public interface can evolve without placing the collections themselves at risk.

The earlier Gateway demonstrated that UNR can create an innovative and valuable statewide geospatial service. Its eventual decommissioning also demonstrated that technical capability at launch does not guarantee long-term viability. Software must be upgraded, vulnerabilities must be remediated, infrastructure must be monitored, metadata must be maintained, and staff must have the time and authority to respond when circumstances change.

The recommendations that follow turn that lesson into an architectural principle: the future repository should remain useful even when one of its applications, vendors, services, or technical components is replaced.

Recommended Architecture

UNR should pursue a phased, hybrid, object-storage-first, and API-first architecture built upon independently managed layers:

1. **Preservation:** Original files, documentation, checksums, provenance, and preservation metadata remain in institutionally controlled preservation storage.

2. **Discovery metadata:** Validated OpenGeoMetadata Aardvark records provide a portable discovery representation while authoritative source and supplemental metadata are retained.
3. **Access assets:** Reproducible derivatives—including COG, GeoPackage, GeoParquet, FlatGeobuf, and packaged original downloads—support analysis and reuse.
4. **Presentation assets:** IIIF, preview COGs, PMTiles, thumbnails, and static maps provide efficient browser display without requiring a map server for every request.
5. **Shared discovery services:** An OpenGeoMetadata-compatible API indexes and exposes the collection through documented web, OGC, QGIS, command-line, citation, and emerging machine-agent interfaces.
6. **Public application:** A branded, accessible frontend provides search, spatial discovery, collection context, previews, downloads, and reuse options.
7. **Operational services:** Monitoring, analytics, security controls, documentation, testing, backups, and maintenance workflows support the system throughout its lifecycle.

Separating these layers allows each to change at a different pace. The public interface can be redesigned without rewriting the metadata. The search service can be replaced without migrating preservation files. New asset formats can be introduced without abandoning the archival originals. This is the foundation of a repository that is both technically adaptable and institutionally sustainable.

Final Recommendations

1. Establish Governance Before Production Procurement

UNR should identify the institutional owners of the repository before committing to a production platform.

At minimum, the program should have a named:

- Executive sponsor
- Service or product owner
- Technical owner
- Metadata and collections owner
- Digital-preservation owner
- Accessibility contact
- Security and privacy contact
- Public-service or user-support contact

These responsibilities may be distributed across University Libraries, campus information technology, GIS programs, and project partners. They should nevertheless be documented. A shared service without explicit ownership can become more difficult to sustain than a locally operated one.

Partnership agreements should explain who may contribute records, who reviews them, which institution preserves contributed assets, how corrections and takedowns are handled, and what happens when a partner leaves the program. Because the repository is statewide in scope, governance should also establish consultation, description, access, correction, and takedown procedures for collections that document Tribal lands, cultural knowledge, archaeological resources, or other sensitive places. UNR should develop these procedures in partnership with affected Tribal Nations and communities before such materials are published.

2. Build the Metadata Foundation Independently of the Geoportal

UNR should adopt OpenGeoMetadata Aardvark as the canonical discovery schema for the future repository. Authoritative source metadata should be preserved, and richer technical, archival, and preservation metadata should remain available as supplemental records.

Metadata crosswalks, controlled vocabularies, normalization rules, validation criteria, and enrichment procedures should be versioned as institutional documentation. Every transformed record should pass structural validation before publication, followed by a separate evaluation of descriptive quality and asset operability.

AI-assisted enrichment may be used to suggest descriptions, subjects, placenames, dates, and other values, but those suggestions should retain provenance and receive appropriate human review. Original values should not be silently overwritten.

This metadata foundation should remain exportable and usable outside the selected geoportal. UNR should be able to move its records into GeoBlacklight, the OpenGeoMetadata API, ArcGIS Hub, an OGC catalog, or a future system without reconstructing the collection from the public website.

3. Separate Preservation, Access, and Presentation Assets

UNR should formally adopt the three-layer asset model described in this report:

- **Preservation assets** retain the original files and associated evidence required for authenticity and long-term stewardship.
- **Access assets** provide standardized, reusable files for download and analysis.
- **Presentation assets** provide lightweight previews optimized for browsers and discovery applications.

Derivatives should be reproducible from the archival originals through version-controlled, documented workflows. Validation should include format conformance, projection and extent checks, HTTP range support, accessibility of linked metadata, and visual inspection in the intended viewer.

Metadata records should connect each asset to its resource through persistent, typed references. The relationship among the original, access copy, preview, thumbnail, and supplemental metadata should never depend upon an undocumented filename convention.

4. Adopt Object Storage as the Default Public-Delivery Layer

Public IIIF, COG, PMTiles, GeoJSON, FlatGeobuf, GeoParquet, thumbnails, and related derivatives should be delivered from S3-compatible object storage through stable HTTPS URLs. A content-delivery network should be used where it improves performance, reliability, or control of outbound traffic.

GeoServer should not be the default preview or distribution mechanism for static collection resources. It should be retained only for requirements that genuinely need server-side processing, such as dynamic reprojection, transactional editing, specialized OGC services, authenticated feature access, raster calculations, or styling that cannot be represented adequately through static derivatives.

UNR should begin with the IIIF, COG, and PMTiles examples already created through OGM Metadata Studio. These prototypes provide a useful starting point for formalizing derivative profiles, validation rules, and metadata references.

This direction aligns UNR with the work underway at Stanford, Princeton, Harvard, and across the broader GeoBlacklight and OpenGeoMetadata communities.

5. Pilot the Shared OpenGeoMetadata API as the Preferred Portal Direction

UNR should first evaluate a shared BTAA/OpenGeoMetadata API with an institutionally branded [ogm-discovery](#) frontend.

This option most closely aligns with the report's metadata, asset, interoperability, reuse, and cost recommendations. It would allow the same collection to support:

- A public geoportal
- Direct API use
- OGC API-Records clients
- QGIS integration
- Command-line tools
- Automated citation
- Collection harvesting
- Research applications
- MCP-compatible and AI-assisted discovery tools

Because this architecture is still maturing, UNR should treat participation as a community partnership rather than a conventional software purchase. Before production adoption, the shared service should have documented governance, export capabilities, service expectations, security responsibilities, and an institutional exit path. UNR should expect to contribute testing, documentation, requirements, and potentially development resources.

The pilot should test representative scanned maps, georeferenced imagery, vector datasets, collection records, compound resources, supplemental metadata, and different rights conditions. Selection should be based on documented acceptance criteria rather than the appearance of a demonstration interface.

6. Retain GeoBlacklight as the Mature Alternative

GeoBlacklight with S3-compatible object storage is the recommended conservative alternative if the shared API service is not sufficiently mature when UNR is ready for production.

GeoBlacklight offers native Aardvark support, an established academic-library community, mature spatial search, and well-understood discovery patterns. It would be an appropriate choice if UNR or a managed hosting partner can provide dependable Ruby on Rails and Solr expertise.

A GeoBlacklight implementation would not require a new GeoServer deployment. IIIF, COG, PMTiles, static thumbnails, and packaged downloads can provide the primary preview and access layer.

ArcGIS Hub Basic should remain a conditional option if it is already included in UNR's educational agreement and a central GIS unit agrees to own and operate the service. UNR should not select it solely on the apparent cost of the license; representative Aardvark metadata, historical maps, cloud-optimized formats, archival relationships, and lossless metadata export should be tested first.

UNR should not establish a new ArcGIS Enterprise, GeoServer, or CKAN deployment solely for this repository.

7. Adopt a Hybrid Hosting Model

UNR should place each repository function in the environment best suited to its institutional and technical requirements:

- Archival originals remain in managed preservation storage.
- Canonical metadata remains under institutional or trusted community control.
- Public derivatives are delivered from S3-compatible object storage.
- The public frontend is distributed through a content-delivery network.
- Application and API services run in a managed or public-cloud environment.
- Administrative functions use institutional authentication.

- Restricted resources remain behind appropriate institutional controls.
- Dynamic geospatial services are introduced only for documented use cases.

The Libraries should avoid purchasing and operating new local hardware solely for the geoportal. On-premises services are appropriate only where UNR already has a supported institutional environment and the repository can enter that environment without becoming an exception dependent upon one person or unit.

8. Make Production Readiness Part of the Acceptance Criteria

Accessibility, security, discoverability, and observability should be requirements for launch—not later enhancements.

The production service should:

- Target WCAG 2.2 AA.
- Provide non-map alternatives for essential search and access workflows.
- Publish an Accessibility Conformance Report.
- Use canonical URLs, XML sitemaps, structured metadata, and social-media cards.
- Protect administrative and high-risk interactions with appropriate WAF and rate-limit controls.
- Avoid challenge pages on public metadata, assets, APIs, sitemaps, and resource pages.
- Require code review, automated tests, linting, dependency updates, secret scanning, and static analysis.
- Monitor the frontend, API, search service, background jobs, and representative assets.
- Measure zero-result searches, failed assets, slow queries, impressions, and downloads.
- Document analytics providers, retention periods, cookies, and information shared with external viewers.
- Provide accessible and accurate error, withdrawal, and maintenance responses.
- Maintain persistent identifiers independently of the current frontend.
- Publish documentation for users, contributors, administrators, and developers.

A failed map preview should not prevent users from viewing the resource's title, metadata, rights, citation, or download links. The system should degrade predictably when individual components are unavailable.

9. Treat the Prototype as a Decision Gate, Not a Production Launch

The Phase 1 prototype should remain intentionally limited. Its purpose is to test the recommended direction, document the architecture, expose gaps, and improve the accuracy of implementation estimates.

The prototype should demonstrate:

- Ingestion of validated Aardvark records

- Delivery of representative IIIF, COG, and PMTiles assets
- Textual, faceted, and spatial search
- Resource and collection pages
- Metadata and asset validation
- Persistent identifiers and typed references
- Citation and download functions
- API and OGC representations
- At least one external reuse workflow
- Basic accessibility and performance testing
- Repeatable installation and deployment
- Documented reindexing, synchronization, and recovery procedures

At the conclusion of the prototype, UNR should conduct a formal readiness review. Production implementation should proceed only if the proposed service has confirmed ownership, a sustainable operating budget, an acceptable accessibility and security posture, and a documented method for exporting its metadata and assets.

10. Fund the Repository as an Ongoing Program

UNR should establish a sustained funding commitment for collection stewardship, infrastructure, and service maintenance.

For planning purposes, the recommended implementation carries a **five-year provision of \$300,000 in constant 2026 dollars**. This comprises \$110,000 for implementation, \$140,000 for operations at \$28,000 annually, and a \$50,000 contingency equivalent to 20 percent of planned expenditure. The five-year period begins with implementation and includes operating support in the first year.

The Libraries should allocate the first \$15,000 of the implementation budget to a representative pilot. Findings from the collection inventory, metadata conversion, and asset processing should inform the remaining budget and release schedule. These estimates assume predominantly automated collection workflows, limited local customization, and access to the shared OpenGeoMetadata API without a mandatory service fee. Staffing assignments and shared-service arrangements should be confirmed before production implementation.

Staff time represents the largest portion of the proposed investment. Metadata management, derivative processing, accessibility review, technical maintenance, documentation, and partner coordination require assigned capacity, including when existing employees undertake the work. Funding requests should distinguish between allocations of existing staff time and new expenditures for personnel, contractors, infrastructure, or purchased support.

Additional collections and service commitments should carry explicit budgets. More extensive metadata remediation, difficult-source processing, new integrations, or increased publishing activity would require additional implementation effort and continuing support. The expanded scenario in Estimated Costs illustrates how these commitments could increase the five-year

provision to \$650,000; its adoption should depend on identified collection needs and institutional priorities.

UNR should review the operating budget annually, using actual workload, storage growth, and service requirements to guide adjustments. This approach connects funding to measurable responsibilities and supports continuity beyond the initial grant period.

11. Plan for Portability and Eventual Replacement

Every component of the repository should have an exit strategy.

UNR should be able to export its metadata, retrieve its assets, rebuild its search index, regenerate its derivatives, redirect persistent identifiers, and replace the public interface without negotiating the recovery of its own collections from an obsolete application.

The project should maintain:

- Complete metadata exports
- Documented storage inventories
- Checksums and fixity reports
- Versioned transformation scripts
- Infrastructure and deployment documentation
- Dependency inventories
- Backup and restoration procedures
- Reindexing and migration instructions
- Vendor and shared-service exit procedures
- Current administrative and technical contacts

Replaceability should not be treated as evidence of weak commitment to a platform. It is evidence that UNR has retained stewardship of the collection.

Recommended Implementation Sequence

UNR should implement the repository through a series of controlled stages. The sequence below is ordered by dependency rather than by calendar date. Before the pilot begins, UNR should convert it into an operating roadmap that assigns a provisional duration, accountable owner, required inputs, decision date, and completion evidence to each stage.

Foundation

- Establish governance and service ownership.
- Inventory collections, metadata, assets, systems, rights, and identifiers.
- Adopt the Aardvark application profile and local metadata guidance.
- Define preservation, access, and presentation asset profiles.

- Establish persistent identifier and publication-state policies.
- Confirm the role of campus IT, GIS services, preservation systems, and external partners.

Proof of Concept

- Transform a representative metadata sample.
- Generate and validate IIIF, COG, and PMTiles derivatives.
- Load the records into the shared OpenGeoMetadata API.
- Deploy a preliminary UNR-branded frontend.
- Test search, previews, downloads, citations, accessibility, and external reuse.
- Compare the results against a small GeoBlacklight implementation if significant questions remain.
- Refine the five-year cost and staffing model using observed workload.

Production Preparation

- Select the platform and hosting arrangement.
- Document shared-service or vendor responsibilities.
- Establish production and staging environments.
- Implement monitoring, security controls, backups, analytics, and deployment automation.
- Complete accessibility testing and an Accessibility Conformance Report.
- Document metadata, derivative, reindexing, synchronization, and incident-response workflows.
- Confirm recurring funding and named institutional owners.

Collection Migration and Launch

- Migrate prioritized collections incrementally.
- Validate metadata and assets before publication.
- Conduct user testing with both GIS specialists and general audiences.
- Publish collection-level context, rights, accessibility, privacy, and reuse documentation.
- Measure failures, zero-result searches, performance, downloads, and support requests.
- Expand statewide participation only after the initial operating model has proven sustainable.

Production Decision Criteria

UNR should not authorize a public production launch until it can answer yes to the following questions:

- Is there an identified service owner with authority to make operational decisions?

- Are technical, metadata, preservation, accessibility, and support responsibilities assigned?
- Is recurring funding available for at least the initial five-year planning period?
- Can the complete collection metadata be exported in a documented format?
- Can public derivatives be retrieved independently of the geoportal application?
- Can derivatives be regenerated from preserved originals?
- Can the search index be rebuilt without reconstructing records manually?
- Have representative records and assets passed automated and human validation?
- Has the application been tested for accessibility, security, performance, and failure recovery?
- Are monitoring, backup, restoration, and incident-response procedures in place?
- Are partner responsibilities and contribution requirements documented?
- Can UNR replace the public interface without changing its persistent identifiers?

These criteria are deliberately stricter than confirming that the application works in a demonstration. They test whether the repository can continue to function as an institutional service.

Conclusion

UNR has an opportunity to build more than a replacement for the Nevada Geospatial Data Gateway. It can establish a durable statewide framework for preserving, describing, discovering, and reusing Nevada's historical and environmental geospatial record.

The recommended path is intentionally evolutionary. It does not require UNR to predict which web framework, cloud provider, or geospatial format will dominate the next twenty years. It requires the institution to preserve authoritative originals, maintain portable metadata, generate standards-based derivatives, publish stable identifiers, and document how the service operates. Those commitments will allow individual technologies to change without severing public access to the collections.

A shared OpenGeoMetadata API, cloud-optimized assets, S3-compatible object storage, and an accessible branded frontend provide the strongest current direction. GeoBlacklight remains a credible and mature alternative. Commercial GIS services may play supporting roles where they offer clear institutional value. None should become the only viable home for UNR's metadata or digital objects.

The most important recommendation is therefore also the simplest: build the foundation before committing to the façade. If UNR invests in its metadata, assets, workflows, people, and partnerships, it will retain the freedom to improve or replace the geoportal as technologies change. That freedom is not incidental to preservation. It is one of the clearest signs that the repository has been designed for the long term.

The Nevada Historical and Environmental Geospatial Repository should be judged not only by what users can find on its first day, but by whether Nevada's communities can continue to find,

understand, cite, download, and reuse those resources decades later. That is the standard against which the project's architecture, funding, and governance should ultimately be measured.

Appendices

Appendix A. Statement of Work Completeness Matrix

This matrix maps the contracted work to evidence in the report and identifies the remaining acceptance actions.

Deliverable	Status	Evidence in Report	Remaining Action
Technical infrastructure assessment	Complete	Geoportal Options; Cloud-Optimized Assets; Web Application Best Practices; Final Recommendations	UNR acknowledges receipt of the comparative platform and architecture assessment.
Platform and technology-stack recommendation	Complete	Geoportal Options; Prototype; Final Recommendations	Record the preferred pilot architecture and the GeoBlacklight fallback.
Hosting, storage, indexing, and delivery architecture	Complete	Cloud-Optimized Assets; Geoportal Options; Estimated Costs; Final Recommendations	Confirm institutional preservation storage and pilot hosting.
Open-source and proprietary option comparison	Complete	Geoportal Options; Estimated Costs	Confirm whether existing ArcGIS licensing or central GIS services change the assumptions.
Cloud, on-premises, and hybrid hosting evaluation	Complete	Estimated Costs; Final Recommendations	Confirm campus constraints, procurement requirements, and security ownership.
Implementation and five-year operating cost estimates	Complete	Estimated Costs	Replace planning assumptions with a dated collection inventory and obtain current estimates before procurement.

Public pricing or vendor evidence	Complete	Estimated Costs	Record dated public pricing and vendor quotations where procurement requires them.
Metadata standards and interoperability guidance	Complete	Metadata Best Practices; Cloud-Optimized Assets; Prototype	Approve a local Aardvark application profile and validation policy.
Discovery, indexing, interoperability, and preservation guidance	Complete	Metadata Best Practices; Cloud-Optimized Assets; Web Application Best Practices; Final Recommendations	Identify preservation ownership and approve persistent-identifier and publication policies.
Strategic-plan technical consultation	Complete	Introduction; Executive Summary; Final Recommendations	Principal investigator records final review and acceptance.
Meetings and principal-investigator consultation	Complete	Project records	Principal investigator confirms participation and receipt.
Working prototype	Complete	Prototype; ogm-discovery; OGM API; OGM Metadata Studio; edu.unr	Preserve stable prototype references and acceptance evidence.
Prototype workflow diagram	Complete	Prototype, Figure 1	Confirm figure placement, caption, alternative text, and source.
Staffing and operations guidance	Complete	Estimated Costs; Web Application Best Practices; Final Recommendations	Assign UNR owners and verify continuing capacity.
Final feasibility and sustainability review	Complete	Executive Summary; Final Recommendations; this matrix	Principal investigator completes the final strategic-plan review.

Appendix B. Implementation Roadmap

Phase 1 — Institutional decisions and collection inventory. Confirm executive sponsorship, hosting and security ownership, preservation responsibilities, pilot scope, rights and sensitivity review, and a dated inventory of metadata and digital assets.

Phase 2 — Metadata and asset preparation. Adopt a local Aardvark application profile, validate and remediate records, establish persistent identifiers, generate cloud-optimized derivatives, and document publication and takedown workflows.

Phase 3 — Pilot implementation. Deploy the OpenGeoMetadata API and the UNR-themed ogm-discovery frontend; configure indexing, previews, analytics, accessibility checks, monitoring, backups, and security controls.

Phase 4 — Acceptance and launch. Complete user testing, remediate accessibility and performance defects, approve operational documentation, train responsible staff, and record a launch decision.

Phase 5 — Operations and evaluation. Review usage, zero-result searches, costs, incidents, metadata quality, derivative validation, and stakeholder feedback on a published schedule.

Appendix C. Initial Risk Register

Security and dependency risk. Reduce attack surface through static cloud-optimized delivery, automated dependency updates, secret scanning, code analysis, patch ownership, and a documented incident-response process.

Sustainability and staffing risk. Assign named service, metadata, preservation, and security owners; fund recurring maintenance; and cross-train staff so the service does not depend on one person.

Cost-growth risk. Establish storage, egress, indexing, and monitoring budgets with alerts; review actual usage quarterly; and revisit the architecture before scaling statewide ingestion.

Metadata and asset-quality risk. Require schema and asset validation before publication, retain source files, record derivative provenance, and publish correction and takedown procedures.

Rights, privacy, and culturally sensitive content risk. Complete rights review; consult Tribal Nations and affected communities where appropriate; and maintain restriction, correction, and takedown pathways.

Accessibility and usability risk. Test against current WCAG AA criteria with automated and manual checks, include disabled users in acceptance testing, and document conformance and known limitations.

Vendor or platform lock-in risk. Preserve portable OpenGeoMetadata records and cloud-optimized assets, document export procedures, and test recovery without a proprietary application dependency.

Appendix D. Sources, Assumptions, and Methodology

The recommendations synthesize the Statement of Work, the grant's project purpose and statement of need, consultation with project leadership, the working UNR prototypes, OpenGeoMetadata and GeoBlacklight community practice, Geo4LibCamp 2026 notes and presentations, and publicly available technical documentation. Cost figures are planning ranges, not bids. They should be refreshed with a dated collection inventory, campus labor rates, current cloud or vendor pricing, accessibility requirements, security controls, and preservation obligations before procurement or launch approval.

Appendix E. Consultant Qualifications

Eric Larson was engaged as the Geospatial Repository Specialist because this assignment sits at an unusually specific intersection of library technology, geospatial discovery, metadata, application architecture, and open-source sustainability.

Larson brings more than 18 years of experience designing complex web applications and digital-library services. He is a core contributor and release manager for GeoBlacklight and has served for many years as Lead Developer of the Big Ten Academic Alliance Geoportal, a collaborative discovery initiative spanning 17 universities.

His consulting work includes geospatial and repository projects for the American Geographical Society Library at the University of Wisconsin–Milwaukee, Johns Hopkins University, New York University, the University of Massachusetts Amherst, the University of Minnesota, the University of Wisconsin–Madison, and other research-library partners. His technical experience includes Ruby on Rails, Python, JavaScript, Apache Solr, Elasticsearch, Blacklight, GeoBlacklight, metadata-driven discovery systems, repository workflows, APIs, and search architecture.

Larson also co-founded Gimlet, a hosted library service used by more than 800 academic, public, community-college, and special libraries. His Master of Arts in Library and Information Studies adds grounding in discovery, metadata, access, preservation, privacy, and stewardship. This combination of library practice, application development, geospatial infrastructure, and collaborative open-source leadership is directly aligned with the project's needs.

Appendix F. Glossary

Aardvark — The OpenGeoMetadata schema used to describe geospatial resources for discovery and interoperability.

BFF, or Backend for Frontend — An application pattern in which an API is designed to serve the needs of a particular frontend while remaining reusable by other clients.

COG, or Cloud Optimized GeoTIFF — A GeoTIFF structured for efficient partial access over HTTP.

GeoBlacklight — An open-source geospatial discovery application built on Blacklight and Ruby on Rails.

IIIF, or International Image Interoperability Framework — A set of interoperable application programming interfaces for delivering and presenting digital images and related resources.

OGM API — The OpenGeoMetadata API used by the prototype to search and retrieve indexed geospatial-resource records.

PMTiles — A single-file archive format for tiled map data that supports efficient range requests from cloud storage.

WAF, or Web Application Firewall — A service that filters and monitors web traffic to reduce common application attacks and abusive automation.